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COMPUTING
& INFORMATICS

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Land cover segmentation and classification of Satellite Images

1211103024
YAP JACK

BACHELOR DEGREE OF COMPUTER SCIENCE
B.C.S (HONS) DATA SCIENCE
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Name of candidate: Yap Jack

Faculty of Computing & Informatics

Multimedia University

Date: 29: 06: 2025

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ABSTRACT

Satellite imagery plays an important role in urban planning and environmental management, especially in tropical regions like Malaysia, where environmental diversity makes land classification rather complex. The project addresses the imbalance in the DeepEarthMY dataset, which does not have enough representation of urban images. Therefore, this project’s main contribution is enhancing DeepEarthMY with additional annotated images from the Sabah and Sarawak regions to create a more balanced dataset. Furthermore, the analysis showed significant differences in pixel and object counts in the rural and urban domains. Rural areas are dominated by Agriculture and Forest categories, while urban areas are dominated by Buildings and Roads. After that, we trained and evaluated several deep learning models, with EfficientNetV2-B2 achieving the best overall performance, making it best for land cover classification. Further comparison between the EfficientNet and VMamba model families highlighted the strengths of each, with EfficientNetV2-B2 and VMambaV2-Small performing best in their respective families. Additional experiments comparing dataset versions and split strategies revealed that random splits on the earlier version (V1) yielded higher accuracy than the manually split refined version (V2), emphasizing the importance of dataset splitting.

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CHAPTER 1

INTRODUCTION

1.1 Basic Introduction

Land-cover segmentation and classification are vital techniques commonly combined together to extract detailed information from satellite images. Land-cover segmentation divides an image into meaningful segments based on land cover types, such as buildings, roads, forests, and barren land, by grouping pixels with similar characteristics. In contrast, land-cover classification assigns images to predefined categories based on their attributes. In this project, the focus is on classifying the images into rural and urban categories.

These techniques are essential advancements to provide fundamental enhancements for urban planning research as well as environmental studies in combination with agriculture and disaster management. Moreover, correct mapping of land cover is essential for urban development and sustainable resource management together with environmental conservation because it enables the analysis of land-use patterns and ecological change monitoring and policy formation. The segmentation and

classification of land cover faces considerable obstacles when carried out in tropical areas like Malaysia because these regions present diverse environmental elements that affect the identification process.

Numerous spectral and spatial pattern modifications arising from dense forests combined with agricultural regions and urbanized areas reduce the ability to accurately segment and classify satellite data. The combination of differing land cover types together with changing environmental conditions produces spectral signature overlaps that results in classification errors through remote sensing systems.

Therefore, the DeepEarthMY dataset represents a breakthrough in remote sensing technology that enables helpful research about land-cover mapping in tropical areas. Remote sensing models for segmentation and classification benefit substantially from the satellite imagery dataset that includes different Malaysian regions. However, two key restrictions exist in this dataset, like the fact that East Malaysia and urban territories are underrepresented. These gaps decreases the dataset's effectiveness in thoroughly representing Malaysia's diverse terrain which reduces a model's ability to achieve better segmentation and classification results.

1.2 Problem Statement

The Completeness and Consistency of the DeepEarthMY dataset is crucial in tropical region segmentation and classification because they ensure all relevant land cover types are accurately represented and the data remains reliable, allowing the model to learn effectively across diverse and complex environments. Building on this, the goal of this project is to enhance the DeepEarthMY dataset and leverage it to train a model that can improve the accuracy of land-cover segmentation and classification. Specifically, the project will focus on distinguishing between rural and urban areas, accounting for the complexities introduced by diverse vegetation, urban structures, and varying climatic conditions. By addressing these complexities, this project aims to contribute to the development of more reliable remote sensing models for land-cover classification in tropical regions like Malaysia.

1.3 Objective

The primary objectives of our research are:

- To analyze and enhance the DeepEarthMY dataset by adding annotated images.
- To perform classification and further analysis on rural and urban images.

1.4 Project scope

This research will focus on image annotation and data exploration. We will annotate satellite images to enhance the DeepEarthMY dataset. Additionally, the project will analyze rural vs urban areas across different climate zones to explore how climate influences land-use patterns. Then, we apply machine learning to perform land-cover classification between rural and urban satellite images.

1.5 Expected Findings

We expect that the analysis of rural and urban images in the DeepEarthMY dataset will reveal significant differences between them. Moreover, we aim to improve the DeepEarthMY dataset by annotating images and adding them to the DeepEarthMY dataset. We also expect that the proposed model will deliver improved segmentation and classification accuracies for rural and urban land-cover segmentation and classification performances.

1.6 Chapter Organization

The report is organized as follows: Chapter 2 provides an overview of the project, including the problem statement, objectives, scope, and expected findings. Chapter 3 presents the background study and literature review, summarizing relevant research on satellite image segmentation and classification. The theoretical

framework is discussed in Chapter 4, covering essential theories for model development. Chapter 5 details the research methodology, including dataset selection, image annotation techniques, model design, and evaluation metrics. Chapter 6 highlights the implementation process, challenges encountered, and solutions applied. In Chapter 7, we analyze the experimental results, evaluating the model's performance in terms of accuracy, efficiency, and effectiveness. Finally, Chapter 8 concludes the report by summarizing key findings, discussing challenges, and suggesting future research directions.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, we first discuss the evolution of land-cover segmentation and classification methods. Then, we move on to the modern approaches of land-cover classification. Next, we discuss the datasets for land-cover segmentation and classification. Lastly, we talk briefly about the evaluation metrics used in land-cover classification.

2.2 Historical Context and Evolution

2.2.1 Land-cover Segmentation

In this subsection, we will talk about the evolution of land-cover segmentation methods.

2.2.1 (a) Classical Segmentation Methods

Classical satellite image segmentation methods involve thresholding, region growing, edge detection and clustering approaches that depend on pixel intensity or

texture features. The thresholding method performs image segmentation by segregating clearly distinguished regions based on pixel intensity values as described by Otsu et al. (1975). The technique of region-growing developed by Pal & Pal (1993) merging neighboring pixels with similar attributes. The process adds pixels one by one to regions based on matching intensity or texture properties between neighboring pixels to segment homogeneous areas in satellite imagery. The Canny edge detector by Canny (1986) serves as a method used for detecting edges which represent region boundaries. Satellite image segmentation at large scales receives robust outcomes from K-means clustering according to Jain (2010) because this method clusters pixels based on their spectral features and spatial distribution. The initial research creates problems with sophisticated datasets because the datasets require modern machine learning or deep learning methods that fit current applications.

2.2.1 (b) Co-Segmentation Methods

The co-segmentation methods employed in satellite image analysis work by performing simultaneous segmentations of multiple images to find matching objects and regions between them. Through this method, the shared information between images produces better segmentation results. Cai et al. (2012) developed an aircraft segmentation technique which used image similarities to enhance detection precision in high-resolution satellite images. Averbuch-Elor et al. (2018) presented co-segmentation

for space-time co-located image collections by expanding foreground and background models throughout the entire collection to handle diverse appearances of common foreground objects. The methods indicate co-segmentation acts as a successful approach to enhance object detection precision and operational stability in satellite image analysis.

2.2.1 (c) Semantic Segmentation Based on Deep Learning

Semantic segmentation is a computer vision problem for which algorithms assign class labels to every single pixel in images to produce advanced scene understanding. Convolutional neural network (CNN) architectures developed by deep learning have revolutionized this field. Long et al. (2015) presented the Fully Convolutional Network (FCN) as the first model to convert existing CNNs into pixel-wise prediction through convolutional layer replacements which enabled end-to-end semantic segmentation training. Ronneberger et al. (2015) developed the U-Net architecture by adding a symmetric expanding path along with skip connections to match high-resolution features with upsampled outputs thus boosting localization accuracy. The DeepLab series by Chen et al. (2017) introduced atrous (dilated) convolutions for obtaining multi-scale context information without adding computational complexity while using conditional random fields for precise segment boundary refinement. State-of-the-art results are achieved across various applications

because deep learning-based approaches have established themselves as fundamental methods in semantic segmentation.

2.2.2 *Land-cover Classification*

In this subsection, we will talk about the evolution of land-cover classification methods.

2.2.2 (a) *Classical Classification Methods*

Classical satellite image classification techniques are considered pixel-based classification and they are known as essential methods for remote sensing. The method classifies images into different land cover types, such as water, forest, and building by using supervised and unsupervised methods. Supervised classification approaches, like Maximum Likelihood Classification (MLC), assign pixels through probability density functions that assume normal spectral distributions according to Lillesand et al. (2015). Unsupervised techniques, like The ISODATA algorithm, execute repetitive pixel grouping based on spectral similarity without requiring any prior information (Abbas et al. (2016)). while classical methods were widely used, advanced classification methods still emerged because traditional methods applied on mixed pixels and complex landscapes create challenges during classification.

2.2.2 (b) Object-Oriented Classification (OOC)

Remote sensing technology utilizes Object-Oriented Classification (OOC) to segment satellite images into objects based on pixels with similar spectral and spatial properties. Then, It groups the objects into established class groups. This method acts as a solution to the issues of pixel-based classification where it struggles with with spatial context understanding and mixed pixels of an image. OOC achieves better accuracy by understanding the spatial distribution patterns of pixels within their environment. The research of Zhang et al. (2019) concluded that Object-Oriented Classification proved to be better than Pixel-Based approaches in terms of accuracy; the researchers examined this via urban feature extraction using GF-2 remote-sensing images. This research confirms that OOC provides exceptional capabilities for accurate satellite image classification of predefined classes.

2.2.2 (c) Satellite Image Classification Based on Deep Learning

Predefined class classification of satellite imagery has reached new heights through deep learning methods. Deep learning models provide better accuracy and efficiency than conventional methods. The classification of satellite images benefits from the effectiveness of Convolutional Neural Networks (CNNs). For instance, Pritt & Chern (2017) created a deep learning system which identifies objects along with facilities within high-resolution multi-spectral satellite imagery through 63 different

class types reaching 83% accuracy. Helber et al. (2019) developed several CNNs that carried out successfully classifications with a 98.57% accuracy on the dataset that Helber created (EuroSat). It can be deduced that deep learning techniques demonstrate their capability to automate satellite image classification tasks while increasing their accuracy according to research findings.

Table 2.1: Historical Context and Evolution Summary

Methods	Disadvantages	Advantages
Classical Segmentation Methods	Sensitive to noise, Difficult to handle complex images, Poor performance with low contrast images	Well-established techniques, Simple to implement, Suitable for basic segmentation tasks
Co-Segmentation Methods	Requires prior knowledge of images, Computationally expensive, May not generalize well to unseen data	Effective for extracting common regions from multiple images, Improves segmentation accuracy by utilizing multiple data sources
Semantic Segmentation Based on DL	Requires large datasets for training, Computationally intensive, May require fine-tuning for specific applications	High accuracy and scalability, Capable of handling complex and large-scale data, Flexible with various architectures (e.g., U-Net, DeepLab)
Classical Classification Methods	Sensitive to noise and outliers, Require a significant amount of training data, May not perform well with complex datasets	Well-suited for small datasets, Simple and fast to implement, Provide interpretable results
Object-Oriented Classification	Requires accurate object delineation, More computationally intensive than pixel-based methods, May struggle with fine-grained classification	Considers spatial relationships for improved accuracy, Effective for complex image scenes, Reduces noise and enhances segmentation
Satellite Image Classification Based on DL	Requires large labeled datasets for training, High computational cost, Prone to overfitting without proper regularization	High classification accuracy, Can handle large and complex datasets, Adapts well to diverse image characteristics (e.g., textures, shapes)

Several image segmentation and classification methods have been

summarized in Table 2.3 along with their respective advantages and disadvantages. It reveals that classical segmentation along with classical classification offer straightforward implementation yet find difficulties processing complicated and noisy satellite image datasets. It also states that co-segmentation and object-oriented classification enhances accuracy through multiple images or spatial relationships but they demand greater computational power. The deep learning methodologies of semantic segmentation and satellite image classification achieve high accuracies and better scalabilities by processing intricate and large datasets, but they demand major computational power and large quantities of labeled data. The table demonstrates how land-cover segmentation and classification methods evolved to produce better performance.

2.3 Modern Approaches of Land-cover Classification

2.3.1 Convolutional Neural Network (CNN)

CNNs are feedforward networks that incorporate convolutional layers, grouping layers, and fully connected layers. A CNN process an input image through several layers. they have wide applications in the field of satellite image classification due to their superior capability to effectively process and analyze image data. There are several versions of CNN such as AlexNet, ResNet, ResNeXt, DenseNet, and EfficientNetV2.

2.3.1 (a) *AlexNet*

AlexNet, developed by Krizhevsky et al. (2012), incorporated three key technical elements which included Rectified Linear Units (ReLU) for accelerated training, dropout for regularization, and cross-GPU parallelization to process the ImageNet dataset. This model marked the first CNN to obtain major progress in the ImageNet Large Scale Visual Recognition Challenge through its combination of 8 total layers including 5 convolutional and 3 fully connected layers. However, large parameters along with heavy computational requirements make AlexNet inefficient compared to modern architectures, thus restricting its ability to scale for complex tasks.

2.3.1 (b) *Residual Network (ResNet)*

In 2015, He et al. (2016) produced ResNet, which became a major breakthrough because the deep neural network was designed to solve problems of disappearance. Through its residual connections Figure 2.1, gradients can directly pass through shortcut paths which enables the successful training of very deep networks. The network architecture utilizes several layers which form residual blocks that optimize computational efficiency and accuracy simultaneously. The ResNet variants exist in five different depths which correspond to their names starting: ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152. The ImageNet benchmarks show that ResNet delivered superior results while showing high flexibility for detecting objects

and segmenting images in addition to its classification capabilities. The performance achievements of ResNet are limited by its high computational and memory requirements which restrict its applications in constrained environments.

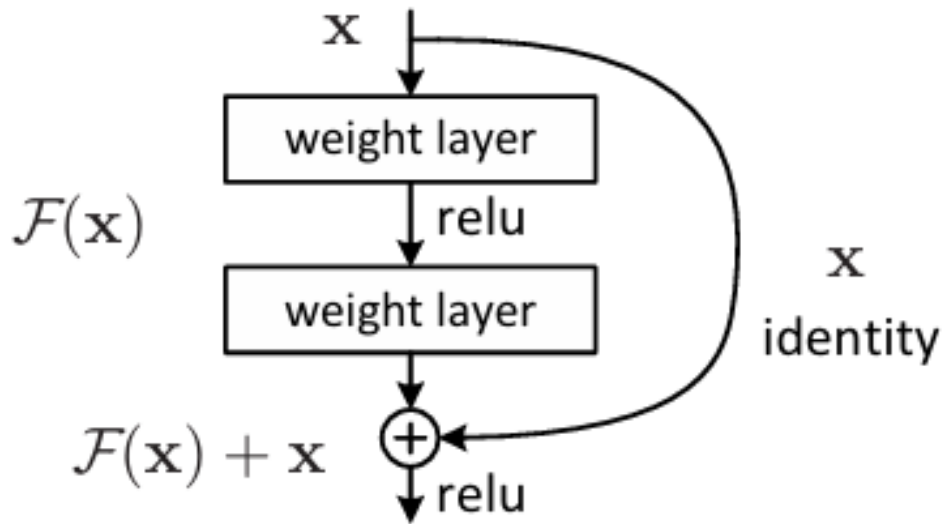


Figure 2.1: Residual Connection (He et al. 2016)

2.3.1 (c) *ResNeXt*

ResNeXt was proposed by Xie et al. (2017) in the year 2017. The aim was to improve both the efficiency and the scalability of deep neural networks. It has cardinality incorporated into the model, which allowed the model to explore a number of parallel paths to learn different feature representations as shown in Figure 2.2. Besides, its grouped convolutions based modular structure introduces a significant improvement in performance over ResNet with lesser parameters and computational costs. The architecture of ResNeXt allows it to be adaptable for tasks in computer vision; this allows a balance of accuracy and efficiency. However, it is still too complex to be

implemented and optimized in certain hardware environments, particularly on devices lacking support for grouped convolutions.

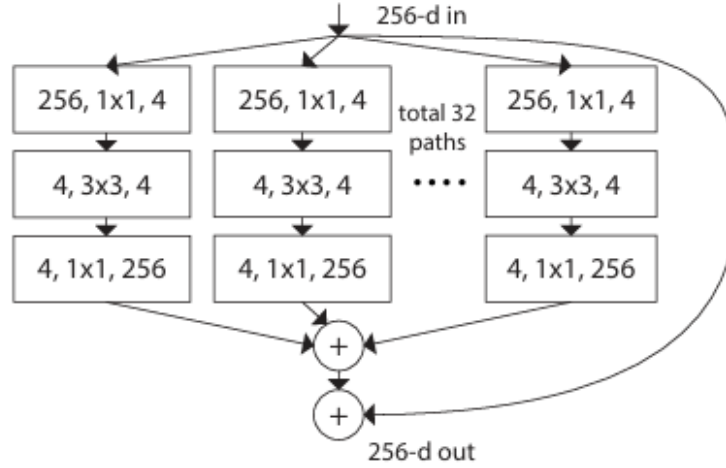


Figure 2.2: A block of ResNeXt with cardinality = 32 (Xie et al. 2017)

2.3.1 (d) DenseNet

DenseNet, created by Huang et al. (2017), introduced a new neural network connectivity. In DenseNet Figure 2.3, every layer is directly connected to all subsequent layers, which encourages feature reuses and diminishes the vanishing gradient problem.

Such a densely connected structure reduces redundancy in feature maps, making the model highly parameter-efficient and easier to train compared to previous architectures. With fewer parameters and less computation, DenseNet achieves competitive performance on benchmarks like ImageNet. However, its nature of dense connections in general increased memory consumption and it is computationally rigorous during inference; hence, its deployment in resource-constrained settings is

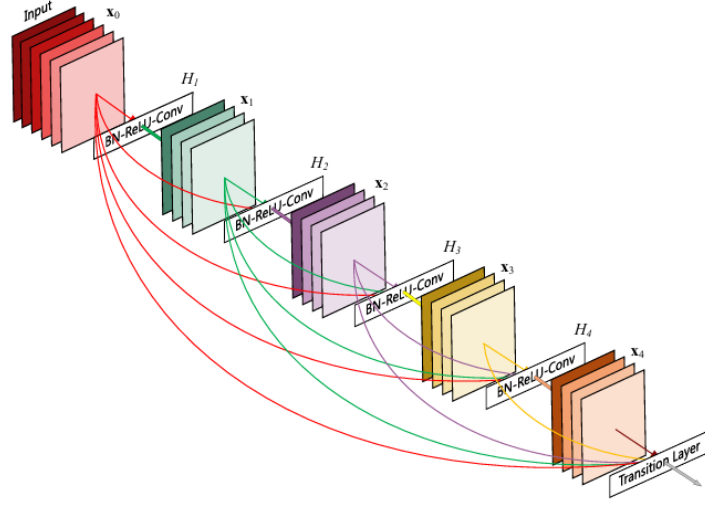


Figure 2.3: DenseNet Architecture (Huang et al. 2017)

limited.

2.3.1 (e) *EfficientNetv2*

EfficientNetV2 was developed by Tan & Le (2021) as an improved version of the original EfficientNet. This architecture used the compound scaling method to upscale model depth, width, and resolution in a balanced manner. It also produced higher accuracy while shrinking computational costs. EfficientNetV2 also introduced more efficient training strategies by employing swish activation functions and fused MBConv blocks, which can accelerate training and improve efficiency. Compared to earlier models, EfficientNetV2 gained significant attention due to achieving state-of-the-art performance with fewer parameters and lesser computational cost. Despite its advances, it is not straightforward to implement, since careful fine-tuning

of the hyperparameters is required; it may also have memory and computation limits in some real-time applications, particularly on mobile or embedded devices.

2.3.2 Transformer

Transformers were originally developed for natural language processing, but they have become very popular in satellite image classification, owing to their ability to capture long-range dependencies with a self-attention mechanism. This allows them to capture long-range dependencies through self-attention. Unlike CNNs, transformers model the global relationships of the image with very high accuracy but they require large datasets and more computation.

2.3.2 (a) ViT

The Vision Transformer (ViT) model is a modified version of the standard Transformer architecture for image classification problems. In ViT, a 2D image is divided into fixed-size patches as shown in Figure 2.4. Then, each patch is flattened and projected into a latent vector space using a trainable linear projection. In order to retain spatial information, these patch embeddings are augmented with learnable positional embeddings. In addition, a classification token similar to BERT's [class] token is prepended to the sequence and the token's state after the Transformer encoder is used for classification. (Alexey (2020))

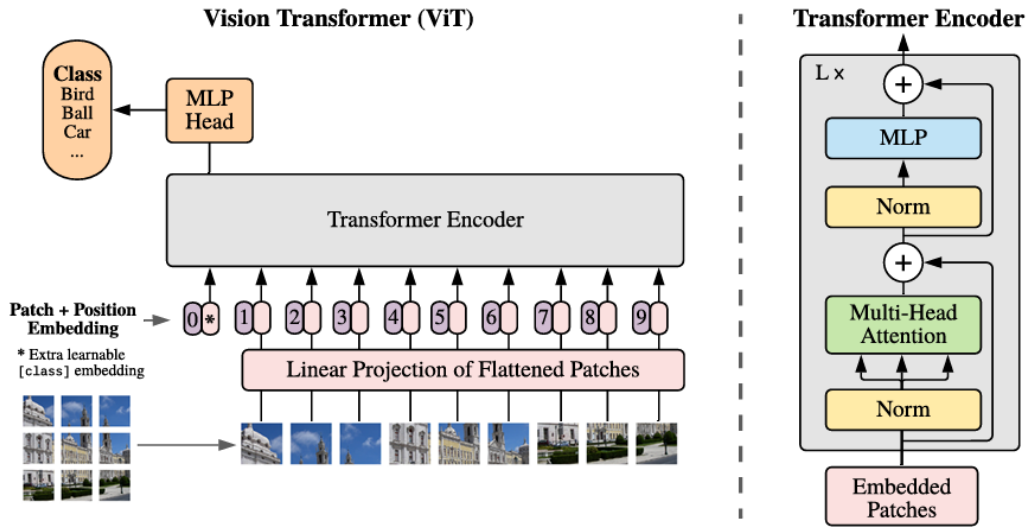


Figure 2.4: Vision Transformer (ViT) overview (Alexey 2020)

The Transformer encoder has layers of multi-headed self-attention (MSA) and MLP blocks where layer normalization (LN) is used before each block. In addition, residual connections are used after each block. The MLP uses two layers and a GELU activation function. In the training process, the output of the classification token is fed into a classification head. The classification head is implemented as an MLP with a hidden layer during pre-training and as a single linear layer during fine-tuning. (Alexey (2020))

The ViT model shows promising results. However, there are still a few challenges. Firstly, it faces challenges in tasks like object detection and segmentation. Tu et al. (2022) states that without extensive pre-training, ViT underperforms on image recognition. There is also a significant gap between self-supervised and

large-scale supervised pre-training. Additionally, further scaling of ViT could improve performance.

2.3.2 (b) *MaxViT*

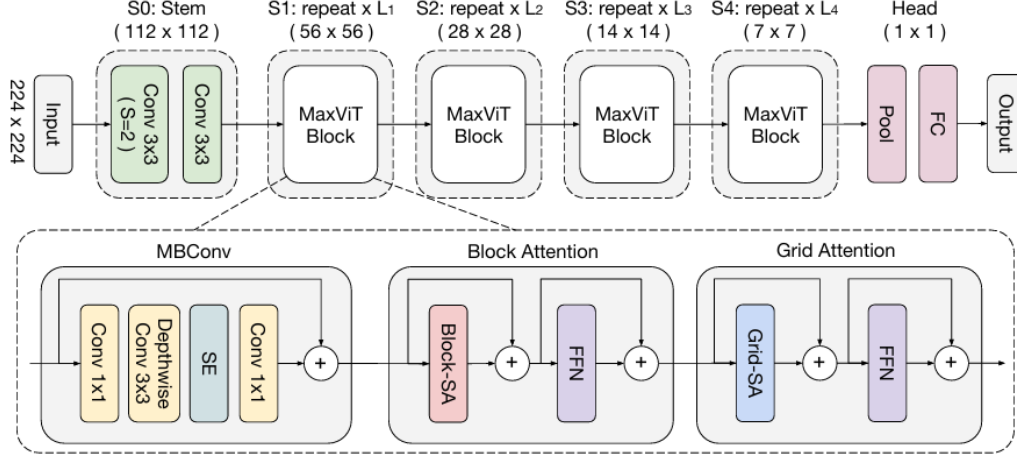


Figure 2.5: MaxViT architecture (Tu et al. 2022)

MaxViT (Multi-Axis Vision Transformer) is a novel vision transformer architecture and it outperforms ViT. It introduces the Max-SA (multi-axis self-attention) module to improve efficiency and performance in vision tasks. It also combines the Max-SA module with MobileNetV2 convolutions (MBConv) in a hierarchical structure. This design allows for efficient global and local receptive fields across all network stages. Therefore, it improves model capacity and generalization abilities. The Max-SA module decomposes the traditional dense self-attention mechanism into two sparse attention forms: blocked attention (local attention) and grid attention (global attention) as shown in Figure 2.5. This significantly reduces the computational complexity from quadratic to linear. Moreover, this approach retains global interactions, which

is a key advantage of self-attention over local convolutions, while making the model computationally efficient. (Tu et al. (2022))

2.3.3 *Mamba*

Mamba uses a neural network structure known as a Selective State Space Model (SSM). SSM was designed to be faster and more flexible than transformers in handling sequence modeling jobs. Mamba uses a different approach than transformers, since it uses a state space to model longer sequences efficiently. It only updates important memories so it can go through long sequences using little computation. Because it works well on tasks related computer vision, Mamba is a promising architecture for designing efficient deep learning models for satellite image classification. (Gu & Dao (2023))

2.3.3 (a) *VMamba*

VMamba takes advantage of the Mamba architecture. It is designed to bring the efficiency and scalability of state space models (SSMs) to computer vision tasks. Unlike most traditional vision models, VMamba uses the state space mechanism from Mamba to capture spatial and long-distance information. By replacing self-attention with input-dependent state updates, VMamba achieves linear computational complexity while maintaining competitive performance across various visual benchmarks. Its

design integrates hierarchical processing and patch-wise representation, enabling it to scale effectively to high-resolution images and long-range visual contexts. This means VMamba can be used to design vision systems that perform fast with minimal memory and it will be able to generalize well (Liu et al. (2024)).

2.3.3 (b) *MambaVision*

MambaVision, developed by Hatamizadeh & Kautz (2024), is a new transformer-based architecture that combines both the strengths of the Mamba framework and Transformer networks, hence providing a hybrid infrastructure able to synthesize benefits from both paradigms. Concretely, this model is designed to enhance performance for such tasks as image classification and feature extraction using the spatial-spectral characteristics of the input images via the Mamba model, while accounting for long-range dependencies that Transformer has shown great capabilities in. MambaVision performs better than standard CNNs by effectively capturing both local and global context information from high-resolution images. Although it is efficient, nonetheless, it presents some challenges due to its complexity in computation.

2.3.4 *Deep Learning Models (with Transfer Learning)*

Transfer learning is a technique in machine learning that leverages existing knowledge from a source domain to solve problems in a related target domain. This

approach is useful when there is a limited amount of labeled data available for the target task. By utilizing pre-trained models on large, well-labeled datasets like ImageNet, transfer learning reduces the need for extensive labeled data in the target domain. It also helps achieve better performance on classification problems. Transfer learning addresses challenges like data scarcity and feature extraction. The process involves transferring knowledge learned from a source dataset to a target dataset (Shi et al. (2021)).

Rangel et al. (2024) utilized transfer learning by fine-tuning the models with pre-trained weights. These weights were sourced from models previously trained on the ImageNet dataset. The experiment results on the EuroSat dataset shown in Figure 2.6 demonstrate the significant impact of using transfer learning. from the results, we can see that the transfer learning employed improved the accuracy of the respective models. This highlights the advantage of transfer learning in scenarios with limited data or when domain-specific training is computationally intensive.

Model name	Accuracy	Accuracy pre-trained
AlexNet	0.837	0.916
ResNet	0.835	0.947
ResNeXt	0.843	0.981
DenseNet	0.925	0.949
MobileNetV3	0.877	0.958
EfficientNetV2	0.908	0.973
ConvNeXt	0.783	0.986
ViT32	0.917	0.972
SwinB	0.926	0.987
MaxViT	0.973	0.990

Figure 2.6: Classification accuracy comparison (Rangel et al. 2024)

Table 2.2: Modern Approaches: Their Strengths and Limitations

source	Proposed Method	Strengths	Limitations
Krizhevsky et al. (2012)	AlexNet	Faster training, avoids overfitting, handles large datasets	Large parameters, high computational requirements, limited scalability
He et al. (2016)	ResNet	Overcomes vanishing gradient problem, supports deeper networks, better generalization	Still complex for some hardware, high computational cost for very deep networks
Xie et al. (2017)	ResNeXt	Efficient with fewer parameters, reduces complexity while maintaining performance	High memory usage, computationally expensive when scaled
Huang et al. (2017)	DenseNet	Highly parameter-efficient, alleviates vanishing gradient, improves training with fewer parameters	Requires massive computational resources, challenging on resource-limited systems
Tan & Le (2021)	EfficientNetV2	Balanced accuracy and efficiency, significantly less computational cost, fast training	Difficult to implement, requires tuning to achieve optimal performance
Alexey (2020)	ViT	Efficient training, superior accuracy with fewer computational costs, suitable for large datasets	Struggles with smaller datasets, computationally expensive compared to CNNs
Tu et al. (2022)	MaxViT	Maintains global interactions, improves	Still very computationally

It can be seen from Table 2.3 that AlexNet and ResNet addressed issues such as training speed and gradient problems, but were computationally demanding. Then, models such as ResNeXt and DenseNet offered better efficiency with fewer resources. Furthermore, transformer-based models like ViT and MaxViT further improved the accuracy, albeit with higher computational costs. MambaVision combines the benefits of Mamba framework and Transformers, which enhances performance but comes with high computational cost compared to CNN-based models. Lastly, pre-trained CNN and transformer models, attained the highest accuracy, although they are the most computationally intensive.

2.4 Datasets for Land-Cover Segmentation and Classification

Satellite image datasets are crucial for training land cover segmentation and classification models, providing valuable insights into Earth’s environment through remote sensing technologies. In this section, we review several widely used satellite datasets, including DeepEarthMY, LoveDA, MLRSNet, EuroSat, and others. DeepEarthMY, MLRSNet, EuroSat, and LoveDA are emphasized due to their relevance to satellite image classification and land cover analysis across diverse climates. DeepEarthMY focuses on tropical and equatorial regions, while MLRSNet offers large-scale global coverage, EuroSat provides an extensive dataset of European landscapes, and LoveDA is widely used for remote sensing in China. These datasets are particularly useful for evaluating image and dataset characteristics, and reviewing them

contributes to a comprehensive understanding of dataset diversity across geographic settings. However, our project primarily focuses on the DeepEarthMY dataset.

2.4.1 MLRSNet dataset

The MLRSNet dataset is a multi-label high spatial-resolution remote sensing dataset for semantic scene understanding. It encompasses 109,161 labeled RGB images collected from over 100 countries. As shown in Figure 2.7, these images are annotated into 46 broad categories like airplane, airport, bareland, baseball diamond, and basketball court; each image is associated with multiple labels selected from 60 predefined classes. The number of images varies by category, ranging from 1,500 to 3,000. The images are 256×256 pixels in size, with spatial resolutions varying from approximately 10 meters to 0.1 meters (Qi et al. (2020)).

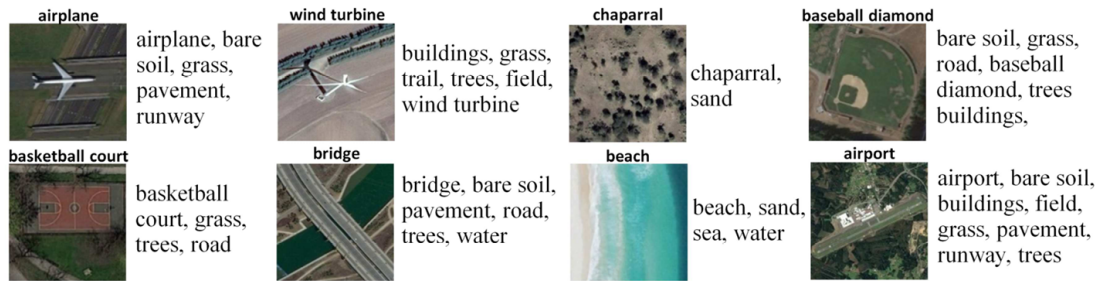


Figure 2.7: Example images of some categories from the MLRSNet dataset (Qi et al. 2020)

This dataset stands out due to its diversity in geographic origin, object categories, hierarchical structure, and multi-label annotations. It also offers large-scale coverage with variations in geographic and seasonal distribution, weather conditions,

and viewpoints, making it an invaluable resource for remote sensing research and applications (Qi et al. (2020)).

2.4.2 EuroSAT dataset

The EuroSAT dataset is an innovative collection of satellite images designed for land use and land cover classification tasks. The dataset contains 27,000 labeled images across 10 distinct land use and land cover classes, each class contains 2000-3000 images. As shown in Figure 2.8, these classes represent diverse environments, including industrial buildings, residential buildings, annual crop, permanent crop, river, sea and lake, herbaceous vegetation, highway, pasture, and forest. The images were acquired from the European Urban Atlas, spanning 34 European countries. Each image has 64x64 pixels with a spatial resolution of up to 10 meters per pixel. All images were manually checked to remove mislabeled images or those affected by snow, ice, or extreme atmospheric effects (Helber et al. (2019)).

Unlike previous datasets, it is multi-spectral, covering 13 spectral bands from the visible, near-infrared, and short-wave infrared regions as shown in Figure 2.9. Furthermore, the dataset is geo-referenced and derived from freely accessible earth observation data, offering a wide array of potential applications. The dataset also contains more classes compared to similar datasets (Helber et al. (2019)).

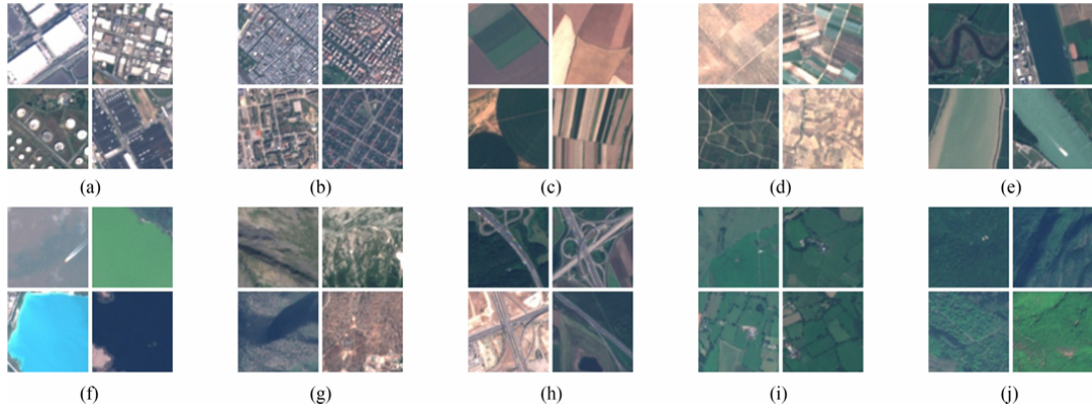


Figure 2.8: Sample images of the classes in the EuroSAT dataset. (a) Industrial buildings. (b) Residential buildings. (c) Annual crop. (d) Permanent crop. (e) River. (f) Sea and lake. (g) Herbaceous vegetation. (h) Highway. (i) Pasture. (j) Forest. (Helber et al. 2019)



Figure 2.9: 13 spectral Bands of EuroSat Images

The EuroSAT dataset is built upon data from ESA's Copernicus program, which manages the Sentinel satellites for Earth observation. Specifically, the dataset uses images from Sentinel-2A and Sentinel-2B, which launched in 2015 and 2017 respectively. These satellites capture global land surface using a multi-spectral imagery (MSI) with 13 spectral bands. Bands B01, B09, and B10 are dedicated to atmospheric correction, while the remaining bands are primarily designed for land use and land cover classification tasks (Helber et al. (2019)).

2.4.3 LoveDA Dataset

The LoveDA dataset is a collection of real urban and rural remote sensing images, designed to advance large-scale land-cover mapping through deep transfer learning methods. At a spatial resolution of 0.3 m per pixel, these images total up to 536.15km² of area across Nanjing, Changzhou, and Wuhan, captured in July 2016 as shown in Figure 2.10. These high-resolution images are sourced from 18 diverse urban and rural scenes across the three cities in China stated earlier. The images also contain multi-scale objects of the same category, which appear in varied geographical landscapes. This introduces significant scale variation for building robust models (Wang et al. (2021)).

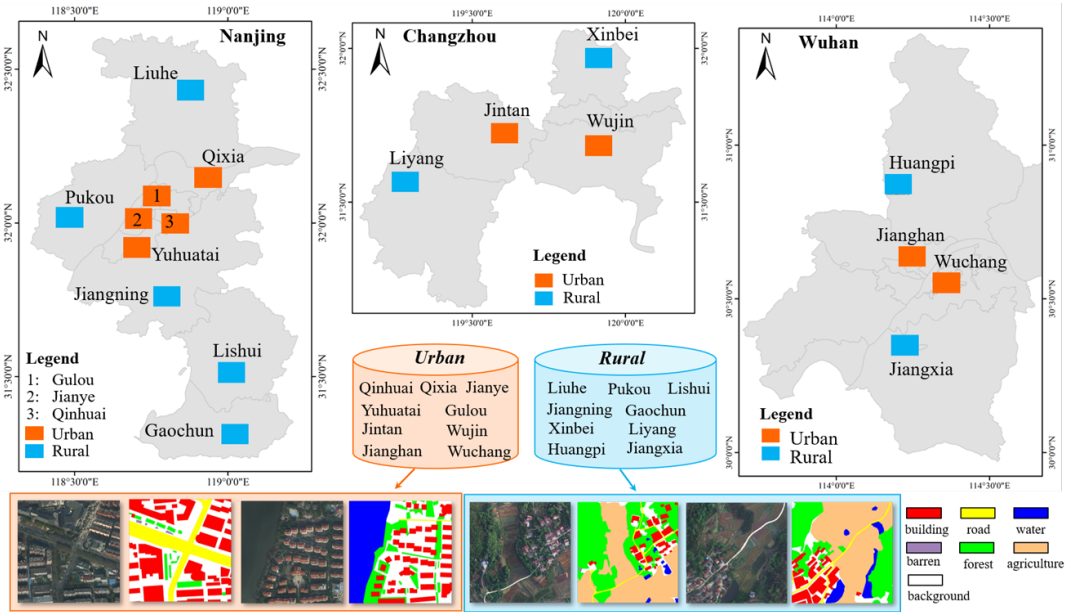


Figure 2.10: Overview of the dataset distribution. (Wang et al. 2021)

Moreover, the dataset includes intricate background details, which is a characteristic of high-resolution images. They present challenges in remote sensing semantic segmentation. The background samples also exhibit rich details and large intra-class variance (Wang et al. (2021)).

In the pixel and object count of the LoveDA dataset, it has the largest number of labeled pixels and land-cover objects compared to other public high spatial resolution (HSR) land-cover datasets as shown in Figure 2.11 (a). In this dataset, urban scenes contribute drastically to the number of buildings as shown in Figure 2.11 (b). Additionally, the background class has the most pixels, consisting of complex samples as shown in Figure 2.11 (c). These complex background samples exhibit greater intra-class variance in intricate scenes, leading to a higher occurrence of false alarms (Wang et al. (2021)).

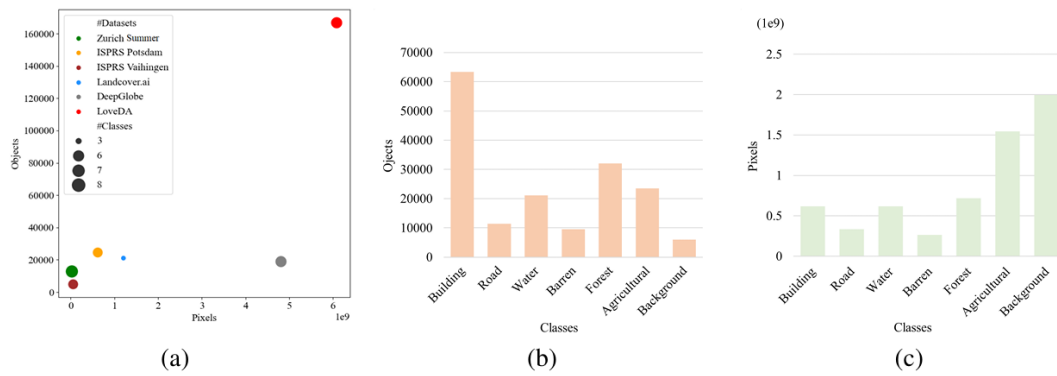


Figure 2.11: Statistics for the pixels and objects in LoveDA dataset. (Wang et al. 2021)

In the statistical analysis, urban areas consist of man-made objects like

buildings and roads due to higher population density, while rural areas have more agricultural land, creating inconsistent class distributions that hinder model generalization as shown in Figure 2.12 (a). Next, rural images have lower standard deviations due to homogeneous landscapes, such as large-scale agriculture and water bodies as shown in Figure 2.12 (b). Lastly, urban buildings have greater size variance, whereas rural buildings are predominantly smaller, reflecting a "long-tail" phenomenon. As shown in Figure 2.12 (c), there are scale differences also between other land cover categories (Wang et al. (2021)).

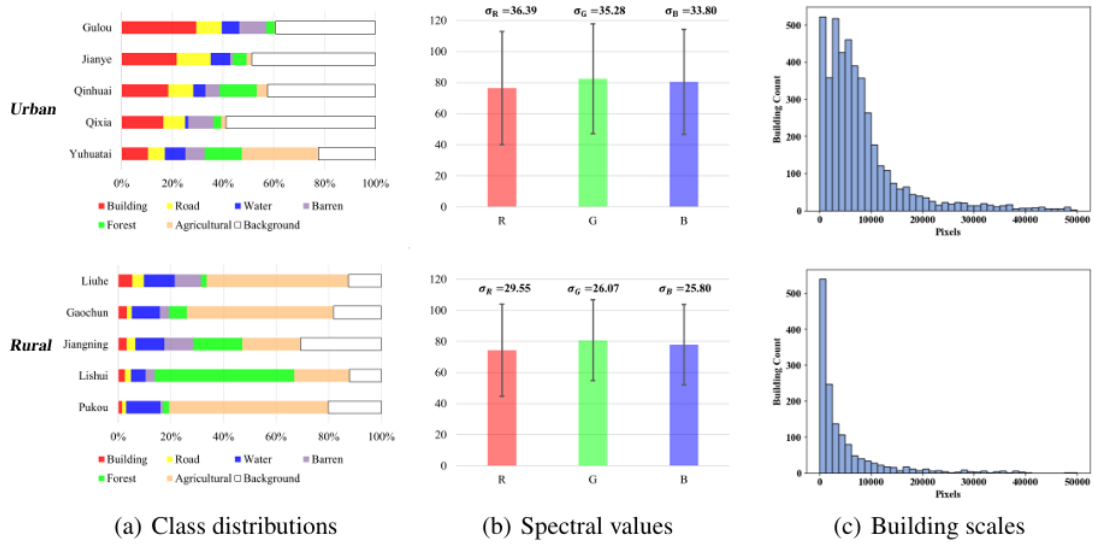


Figure 2.12: Statistics for the urban and rural scenes in Nanjing City. (Wang et al. 2021)

Compared to typical UDA datasets, the LoveDA dataset emphasizes stylistic differences between geographical environments. It is designed for both advanced semantic segmentation and UDA evaluations. Initial experiments have tackled its inherent challenges, paving the way for further research. The dataset calls for robust

representational architectures and UDA methods to enhance large-scale land-cover mapping (Wang et al. (2021)).

2.4.4 *DeepEarthMY Dataset*

DeepEarthMY is a high spatial resolution (HRS) dataset specifically designed for equatorial environments. Covering over 378.15 km², the dataset represents Malaysia's diverse landscapes. The images of the dataset have a resolution of 1024x1024 pixels with 0.3 m per pixel. The dataset is captured using SASPlanet and it includes 4,007 annotated images from 52 diverse regions in Malaysia (Figure 2.13), covering land types like forests, buildings, roads, water bodies, agricultural areas, and barren land (Shaaban et al. (2016)).

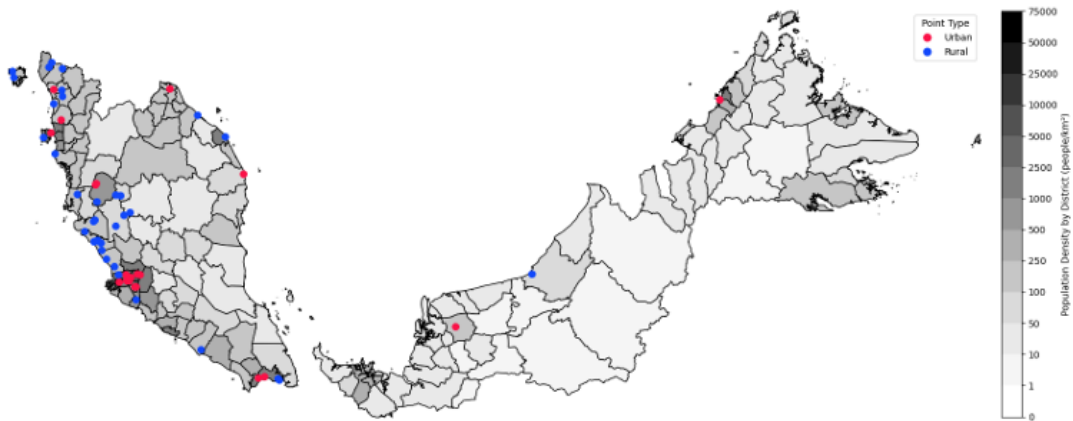


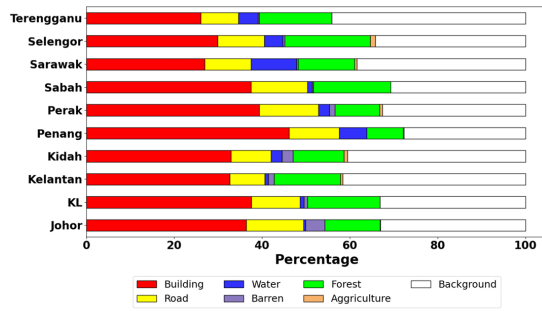
Figure 2.13: Distribution of DeepEarthMY over population density. (Shaaban et al. 2016)

The dataset is systematically curated based on population density within State

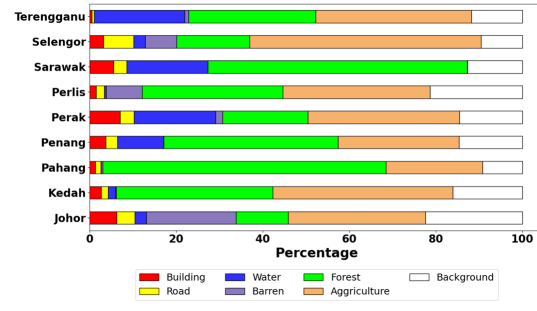
Legislative Assembly (DUN) boundaries. Urban areas are defined as regions with a population density over 1,000 people per km², while rural areas have lower densities. Out of the 4,007 images, 1,402 come from urban areas, and 2,605 are from rural regions. An additional 295 images focusing on water regions (such as rivers and beaches) were collected to enhance the coverage of land-cover categories. This ensures a well-rounded representation of Malaysia's geographic and demographic diversity (Shaaban et al. (2016)).

As shown in Figure 2.14, the analysis reveals land-cover differences between urban and rural regions. Urban areas exhibit higher concentration of infrastructure, with building coverage (26.02% to 46.15%), and road coverage (7.96% to 13.48%). In contrast, rural regions show minimal building (0.48% to 6.99%) and road coverage (0.53% to 6.91%). Water bodies also vary, with rural regions displaying a broader range (0.15% to 20.92%) compared to urban areas (0.49% to 10.39%). Rural areas are dominated by agricultural (0.09% to 53.51%) and forest land (up to 65.45%), whereas urban regions have minimal agricultural (up to 1.12%) and forest cover (8.31% to 19.50%) (Shaaban et al. (2016)).

The urban histogram shows a strong skew towards smaller buildings, with most structures ranging from 0 to around 1,800 m², and a sharp decline in larger



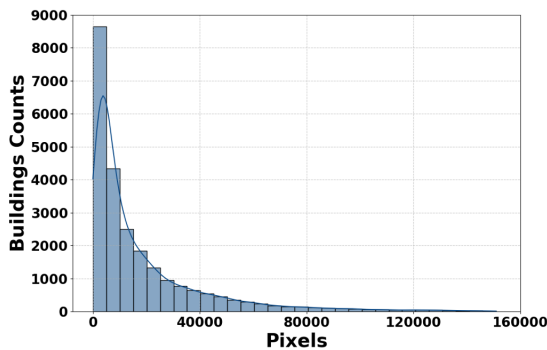
(a) Urban



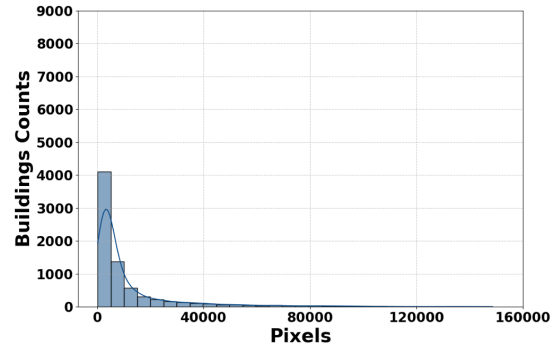
(b) Rural

Figure 2.14: Class Distribution (Shaaban et al. 2016)

buildings (Figure 2.15a). Rural areas also lean towards smaller buildings, but the decline is more gradual, with a wider range of sizes, though large buildings are still rare (Figure 2.15b). Hence, urban areas are characterized by densely packed small buildings, while rural areas show more variation in building size (Shaaban et al. (2016)).



(a) Urban



(b) Rural

Figure 2.15: Building Counts (Shaaban et al. 2016)

The bar chart in Figure 2.16a shows that the background class has the highest pixel count, which suggests that undefined land types are abundant. This is followed by agriculture, forest, and buildings. Roads and water bodies have moderate pixel counts, while barren land has the lowest. This highlights the dominance of natural land types

(Shaaban et al. (2016)).

The bar chart in Figure 2.16b shows that urban areas have higher mean spectral values than rural areas for all channels. Urban areas also show higher standard deviations, indicating greater spectral variability due to diverse infrastructure and land cover. In contrast, rural areas exhibit lower variability with more consistent spectral values, attributed to the prevalence of natural landscapes like forests and agricultural fields (Shaaban et al. (2016)).

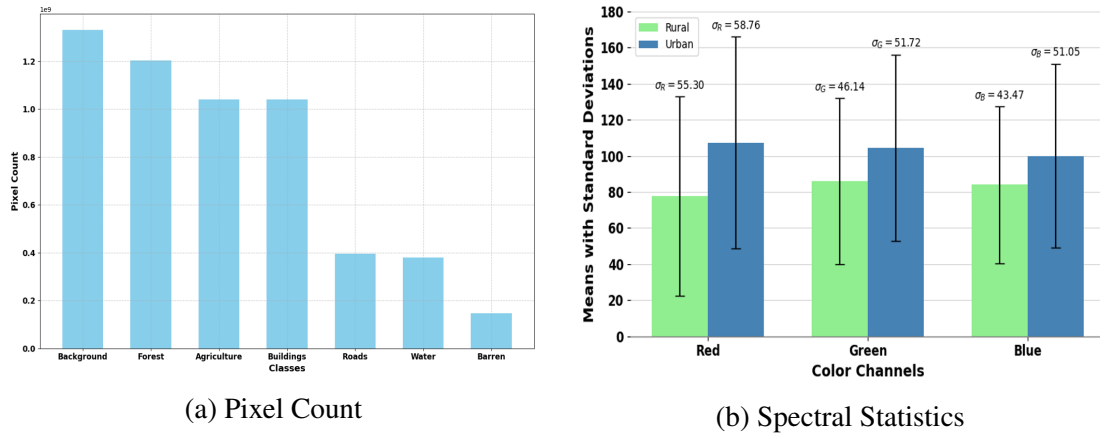


Figure 2.16: Statistics of DeepEarthMY (Shaaban et al. 2016)

While datasets like MLRSNet, EuroSat, LoveDa, and DeepEarthMY have been extensively reviewed for satellite image classification, the following datasets are the ones that I have examined only briefly. In the study of Basu et al. (2015), the DeepSat datasets include SAT-4 and SAT-6, both for satellite image classification tasks. SAT-4 consists of 4 classes, while SAT-6 contains 6 classes of land cover. The images in SAT-4

and SAT-6 are at a resolution of 30 meters. Moreover, the DeepGlobe dataset by Demir et al. (2018) contains 1,146 0.5 meter resolution satellite images of the whole world; the dataset has seven classes and it is suitable for land cover classification. Furthermore, the SpaceNet dataset by Van Etten et al. (2018) consists of 6,000 images at a resolution of 0.5 meters and it focuses on urban land cover in the United States; there are two classes: building and non-building. Finally, LandCoverNet by Alemohammad & Booth (2020) is a large-scale, global benchmark dataset for land cover classification; it includes images in 0.3-meter resolution and a wide range of land cover classes.

After thorough analysis, the DeepEarthMY dataset is found to have various limitations. These limitations include: underrepresentation of urban areas and underrepresentation of East Malaysia.

Table 2.3: Summary of Datasets

Dataset	Source	Classes	Resolution (m)	Image Width	Images	Area (km ²)	Year	Region	Climate
LandCoverNet	Alemohammad & Booth (2020)	7	10	256x256	1980	30000	2020	Global	All
SpaceNet	Van Etten et al. (2018)	2	0.5	1024x1024	6,000	2,544	2017	United States	Warm-Temperate, Polar, Arid
DeepGlobe	Demir et al. (2018)	7	0.5	1024x1024	1,146	1,716.9	2018	Global	All
SAT-6	Basu et al. (2015)	6	1	28x28	405,000	317.52	2015	United States	Warm-Temperate, Polar, Arid
SAT-4	Basu et al. (2015)	4	1	28x28	500,000	392	2015	United States	Warm-Temperate, Polar, Arid
MLRSNet	Qi et al. (2020)	60	10-1	256x256	109,161	722,559	2020	Global	All
EuroSat	Helber et al. (2019)	10	10	64x64	2,700	1,106.33	2015-2017	Europe	Warm-Temperate, Polar
LoveDA	Wang et al. (2021)	7	0.3	1024x1024	5,987	536.15	2021	China	Warm-Temperate
DeepEarthMY (ours)	Shaaban et al. (2016)	7	0.3	1024x1024	4,007	378.15	2023	Malaysia	Equatorial

As shown in Table 2.3, the DeepEarthMY dataset overcomes some of

the limitations of existing land-cover datasets. DeepEarthMY is superior in terms of resolution, region-specific coverage, and timeliness. DeepEarthMY offers high-resolution imagery at 0.3 meters. This allows for more precise land-cover classification. Whereas, datasets like MLRSNet, EuroSat, LandCoverNet, SAT-4, and SAT-6 offer lower resolutions from 1 to 10 meters. Besides that, DeepEarthMY focuses on the rural landscapes in Malaysia and captures the unique features of equatorial climates, which are underrepresented in other datasets. DeepEarthMY was collected in 2023. Therefore, it offers more current and relevant data than older datasets such as EuroSat (2015-2017) and SAT-4 (2017).

CHAPTER 3

THEORETICAL FRAMEWORK

The theoretical framework for our research utilizes advanced integration of deep learning models to address the challenges of land-cover classification in satellite imagery. This study underpins two prominent architectures: ResNet-50 and MambaVision, for land cover classification of satellite imagery. The primary goal is to optimize the performance of these models in extracting meaningful features from satellite images for precise land-cover classification.

3.1 Deep Learning

The deep learning revolutionized satellite image classification through its ability to automatically acquire hierarchical features from raw images. Typically, deep learning models, like convolutional neural networks (CNNs), possess the ability to extract spectral and spatial features from satellite images without manual feature extraction methods, which are used in traditional models. These deep learning models apply activation functions to introduce non-linear properties, allowing them to detect complex data patterns. In addition, they also utilize loss functions to help evaluate classification mistakes and optimize learning by using backpropagation. Experts apply

CNNs especially in land-cover classification applications to achieve accurate terrain recognition. Modern deep learning models, like ResNet and MambaVision, achieve better results in image classification compared to traditional models. These models also extract relevant information together with hidden structures from extensive datasets containing detailed data. These models find numerous applications in remote sensing operations because of their capabilities.

3.1.1 Convolutional Neural Network (CNN)

Convolutional neural networks (CNNs) are a class of deep learning models that capture spatial hierarchies in images via convolutional layers. This is very handy for classifying satellite imagery because CNNs can learn low-level features, such as edges and textures, and high-level features, such as shapes and patterns, which are important for distinguishing various land-cover classes. CNN architecture contains various stacked convolutional layers, pooling layers, and fully connected layers capable of providing exceptional processing of large satellite images.

3.1.1 (a) Residual Network (ResNet)

Residual Network (ResNet) is a CNN-based model and it is especially effective for land-cover classification of satellite imagery because it can capture fine patterns and train deep networks quite efficiently.

The network's first convolutional layer with a 7x7 kernel and stride 2 is used to capture spatial dependencies in large image patches. This feature is particularly beneficial when working with high-resolution satellite images, where both large features such as forest or water bodies and small features must be identified. The capacity of ResNet to handle such varied scales enables it to perform well at detecting both large and small features for different land-cover classes.

Moreover, ResNet includes the use of residual blocks (Figure 3.1), which avoids the vanishing gradient; thus enabling the training of very deep networks that could learn low-level features, like edges, textures, and also higher-level patterns, like shapes and terrain boundaries. This hierarchical feature extraction is extremely important for recognizing various types of land cover such as water, vegetation, or urban areas. More so, the deep architecture of ResNet allows refinement of features through its layers to learn a very subtle spatial and spectral details from the high-resolution satellite images.

ResNet also employs max pooling and global average pooling (GAP) to reduce the spatial dimensions of feature maps. While max pooling summarizes information by selecting the most important features, GAP globally summarizes the spatial features. This dimensionality reduction not only decreases computational

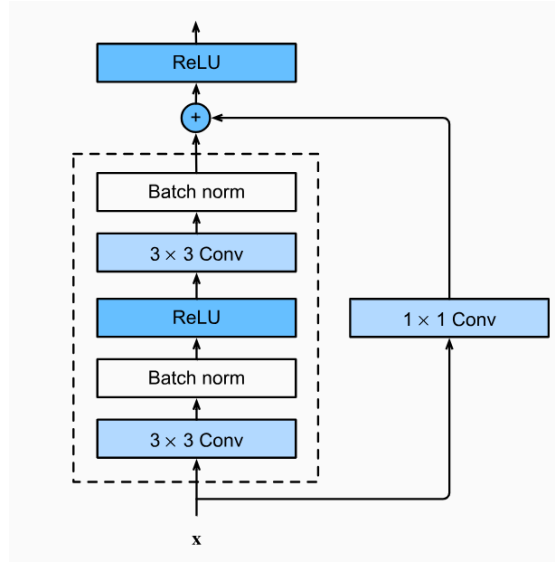


Figure 3.1: Residual Block

requirements but also reduces the risk of overfitting, especially in deep networks. GAP, in particular, converts 2D feature maps into a 1D vector that represents the entire image, enhancing generalization and improving performance on unseen satellite imagery.

The robust architecture of ResNet adjudges accuracy, even with large or complex datasets. Therefore, ResNet is a good candidate for land-cover classification.

3.1.2 *Mamba*

Mamba is a newly proposed deep learning model for fast and accurate image classification. It merges the benefits of convolutional networks and transformers, and local feature extraction and global context awareness are integrated into the model. It generally starts from a convolutional neural network(CNN) for capturing low-level

features to transformer layers that involve self-attention mechanisms to catch long-range dependencies and contextual information. Mamba incorporates this hybrid approach to deal with both fine-grained spatial details and broader environmental patterns. This makes it very effective for complex tasks like land-cover classification in satellite images.

3.1.2 (a) VMamba

VMamba (Figure 3.2) is a novel vision backbone architecture designed for efficient and scalable visual representation learning. It is inspired by the Mamba model from NLP, which is based on State Space Models (SSMs), and adapts its linear-time computation capabilities to vision tasks. Liu et al. (2024)

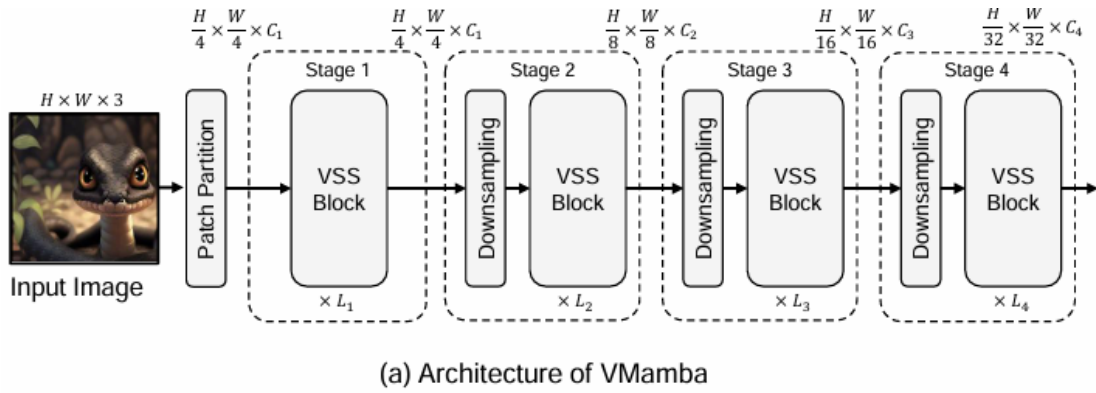


Figure 3.2: Architecture of VMamba Liu et al. (2024)

In the first stage, VMamba begins with a patch partition layer, which converts the input image into a 2D grid of patches for processing. This is done by applying a convolutional layer that splits the image into smaller patches while preserving local

spatial information. The partition reduces the spatial resolution from the original image size of 224×224 pixels to a smaller grid (for example: 56×56 patches), and the patches are represented as a feature map. Following this, the map is processed as a sequence of vectors by a stack of vision state space (VSS) blocks, which leverages state space models. These blocks enable the model to efficiently capture both local and long-range dependencies early on. This stage is crucial for extracting foundational visual features and sets the stage for deeper, more abstract representations in later stages.

In the second stage, it applies downsampling to reduce the spatial resolution of the token sequence, typically by using convolutional layers with strides greater than one. For example, the spatial size is reduced from 56×56 tokens to 28×28 , which decreases computational costs and allows the model to focus on larger and more abstract regions of the image. The resulting feature map passes through the next sequence of VSS blocks. These blocks perform state-space modeling in 2D, enabling the model to capture long-range dependencies and refine the semantic representation at this coarser resolution.

In the third stage, it further downscales the feature map using another convolutional downsampling layer, reducing the resolution from 28×28 to 14×14 . This step continues the model’s hierarchical abstraction, enabling it to concentrate on

more complex structures and relationships across broader areas of the image. Once downsampled, the 14×14 feature map is fed into another stack of VSS blocks. These blocks perform advanced state-space modeling at this mid-level resolution, capturing semantic relationships that span larger portions of the image. As a result, the model can recognize patterns such as object parts or inter-object interactions with improved efficiency and expressiveness.

In the fourth and final stage, one last downsampling operation is performed, shrinking the spatial dimensions from 14×14 to 7×7 . This coarse resolution allows the model to operate at a global level, where the emphasis is on the overall structure and high-level context of the image. At this scale, a final stack of VSS blocks processes the patches, ensuring that the model aggregates and refines global semantic information. The output of this stage serves as the final representation of the image and is typically passed into a classification head or other task-specific layers. This stage is essential for capturing the full understanding at the image level required for tasks such as classification or detection.

CHAPTER 4

RESEACH METHODOLOGY

In the present research project, our purpose is to develop a research methodology for land-cover classification of satellite imagery using deep learning architectures. An in-depth background research has been conducted in an attempt to understand the challenges and methodologies involved in land-cover classification.

To maximize both effectiveness and efficiency in this study, the research methodology is segmented into a range of key phases. These phases include problem and objective formulation for research, satellite image dataset collection, dataset analysis, prototyping, training and testing of the prototype, prototype performance evaluation, annotating additional images to enhance the dataset, preprocessing the dataset, developing and implementing the final model, training and testing of the model, model performance evaluation, model refinement according to the results of evaluation, and, finally, applying model on test case dataset.

By sequentially defining the research process through these discrete phases,

our methodology aims at optimizing and simplifying the pipeline, with each part being performed with careful detail, and in this manner, allowing for the achievement of accurate and reliable land cover classification.

4.1 *Proposed Method and Research Methodology Steps*

The proposed work involves developing deep learning models for satellite imagery land-cover classification with ResNet as the prototype and MambaVision as the final model. Figure 4.1 is a flowchart depicting the methodology adopted in this study. The problem statements and objectives for the project were determined beforehand.

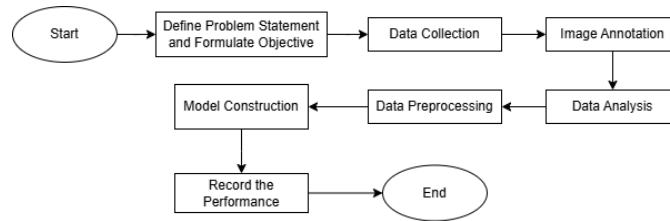


Figure 4.1: Workflow of the Research

The following sections will present an in-depth and thorough analysis of each stage in the research methodology, from collecting the dataset to an overall evaluation of the final model's performance in land-cover classification.

4.2 Image Annotation Plan

We will perform image annotation to correct any imbalances and improve the representation of urban images. We will specifically enhance the amount of urban images to provide a more balanced distribution between rural and urban classes. A well-balanced dataset is required for training a robust classification model, as an imbalanced dataset can result in biased predictions favouring the majority class. To address this, we will improve the dataset by annotating more urban images into the dataset.

4.3 Dataset Analysis

In this project, we will be using the DeepEarthMY V2 dataset, which is DeepEarthMY from Shaaban et al. (2016) with the additional annotated images. The images of the dataset have a resolution of 1024x1024 pixels with 0.3 m per pixel. The dataset consist of rural and urban images. It also contains land types like forests, buildings, roads, water bodies, agricultural areas, and barren land.

The dataset will subsequently be analyzed to examine the rural and urban distribution of land cover types through pixel and object count analysis. By utilizing boxplots and barplots, pixel count analysis will illustrate dominant rural and urban land cover classes, and will identify variation and outliers. Object count analysis will

subsequently assess individual land cover object size and distribution, and will identify rural and urban region variation in structure and distribution. All these will go towards balancing datasets and improving model accuracy for land cover classification.

4.4 Model Construction

Before training the models, we will preprocess the dataset by standardizing the input images, including resizing to a smaller resolution. Then, we split the dataset into training, validation, and testing sets.

VMamba is chosen for the final model because it achieves efficient and scalable vision modeling using state-space models (SSMs) that scale linearly with input size. Unlike traditional attention-based architectures that suffer from quadratic complexity as input resolution increases, VMamba leverages a Visual State Space Model (VSSM) built on top of the VSS blocks. These blocks integrate the 2D Selective Scan (SS2D) module, which preserves the benefits of the original Mamba architecture, such as dynamic weighting and global receptive fields, while effectively adapting it for vision tasks.

Moreover, another reason we chose VMamba is that it offers a more practical balance between performance and efficiency compared to MambaVision. Although MambaVision demonstrates strong accuracy, it is significantly more computationally

intensive, which can hinder scalability and slow down inference in real-world applications. On the other hand, VMamba reduces memory and computational overhead by simplifying the architecture and optimizing the scan strategy for 2D spatial data. This makes it a more suitable choice for large-scale vision tasks where both speed and accuracy are critical.

After implementation of the final model, we will compare it with other models discussed in the literature review. The models we have chosen to compare VMamba with are AlexNet, ResNet50, EfficientNet, EfficientNetV2, and ViT Transformer. These models are selected because they represent key milestones in deep learning evolution, from early CNNs to modern efficient architectures and transformers, allowing us to fairly evaluate VMamba's performance across different generations and types of models.

4.5 Evaluation Metrics

Normally, the next logical step after building each classification model is model evaluation. It helps to identify which of the models perform best in the given dataset and offers insight into how well the final model will generalize into the future. Key metrics include accuracy, precision, recall, and the F1-Score. This helps us perform an analysis about how well or badly the model performs in predicting correctly or mistakenly classifying any instances.

In classification models, there are four possible outcomes: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). TP refers to the number of correctly identified positive samples, while TN indicates the correct identification of negative samples. FP represents the misclassification of positive samples, and FN refers to the misclassification of negative samples. These terms are vital in evaluating model performance.

(i) Accuracy: Accuracy is the proportion of correct predictions out of all predictions made. It is the most commonly used metric to evaluate model performance. The formula for accuracy is. The formula for accuracy is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

(v) Loss: The Loss measures how well the model's predictions match the actual labels, providing a single metric to evaluate model performance, especially during training. It quantifies the error between predicted and true values and is minimized to improve model accuracy:

$$\mathcal{L} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (4.2)$$

(ii) Precision: Precision measures the percentage of correctly predicted instances belonging to a specific class. It highlights the accuracy of positive predictions. It is especially useful when dealing with imbalanced datasets, as accuracy alone can be misleading. The formula for precision is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.3)$$

(iii) Recall: Recall measures the proportion of correctly predicted positive instances out of all actual positive cases. It is useful for evaluating the model's ability to detect positive samples. The formula for recall is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.4)$$

(iv) F1-Score: The F1 Score balances precision and recall, providing a single metric to evaluate how well the model performs, especially in cases of unbalanced datasets. It is calculated as the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.5)$$

CHAPTER 5

IMPLEMENTATION

In this chapter, we will discuss the implementation phase of our satellite image classification system. It outlines the steps taken to develop the satellite image classification system. The purpose of the implementation chapter is to provide a clear picture of how the system was developed. This chapter will include sections such as Development Environment Setup, Data Collection and Preprocessing, Exploratory Data Analysis (EDA), and Model Implementation.

5.1 Development Environment Setup

5.1.1 Tools and Platforms Used

The required tools and platforms used for model construction are shown in Table 5.1

Table 5.1: Tools and Platforms Used

Tool/Platform	Usage
WSL (Windows Subsystem for Linux)	Running and testing models specifically designed for Linux.
Visual Studio Code	Code editor for writing and debugging VMamba code
Jupyter Notebook	For prototyping, data visualization, and testing models

5.1.2 Programming Languages and Libraries Used

The required programming languages and libraries used for model construction are shown in Table 5.2

Table 5.2: Programming Languages and Libraries

Programming Languages/Libraries	Usage
Python	Running and testing models specifically designed for Linux.
numpy	Numerical computations like array and matrix operations
pytorch	Used to run and test models
torch, torchvision.datasets, torchvision.transforms	Loading and preprocessing image datasets.
os, glob, shutil, re	File and directory handling (os, glob, shutil) and regular expressions (re) for filename or path processing.
PIL.Image	Handling and processing image files.
rasterio, rasterio.plot.show	Reading, writing, and visualizing satellite images.
matplotlib.pyplot	Creating visualizations such as plots, graphs, and confusion matrices.
seaborn	Used for statistical data visualization.
tensorflow, tensorflow.keras	building and training neural networks.
tensorflow.keras.layers	Used to construct different layers in neural network architectures.
itertools	Used for creating iterators.

5.1.3 Software Stack and Version Details Used

The required software stack used and version details for model construction are shown in Table 5.3

Table 5.3: Software Stack and Version Details

Software Details	Stack/Version	Usage
CUDA Toolkit 12.9		Used to make training of neural networks faster.
Python 3.10		Used due to library compatibility.
Anaconda Navigator		Managing environments and dependencies
Google Maps		Searching and viewing areas for SASPlanet
SASPlanet		Accessing and downloading satellite imagery and geospatial data
Labelme		Image annotation
Department of Statistics Malaysia (DOSM)		Evaluate the population density of each State Legislative Assembly

5.2 Data Collection and Preprocessing

5.2.1 Data Sources

For this project, we utilized the DeepEarthMY dataset. The dataset provides high-resolution satellite images of various regions in Malaysia. The data files were all kept locally, with images grouped by region type, state, district, and Constituency. The images are stored as png files in methodically arranged folders.

In addition, manual image annotation was done due to the limited number of urban images. We obtained new satellite images from SASPlanet, then labeled each land-cover type of each image using the LabelImg tool. As a result of this process, the urban class became wealthier, and the balance and variety of the dataset improved.

Table 5.4 below presents the complete summary of the annotated areas:

Table 5.4: Urban Constituencies in Sabah and Sarawak with Population Density, Image Count, and Coordinates

State	District	Constituency/State Assembly	Population Density	Images	Coordinate (Upper Left)	Coordinate (Lower Right)
Sabah	Kota Kinabalu	Api-Api	5251	57	5.983520, 116.072190	5.961662, 116.094160
Sabah	Sandakan	Elopura	5339	54	5.873130, 118.085300	5.851270, 118.107270
Sabah	Tawau	Kukusan	3737	62	4.283731, 117.878293	4.261811, 117.900263
Sabah	Sandakan	Sungai Sibuga	2045	64	5.896488, 118.039487	5.874630, 118.061457
Sabah	Putatan	Petagas	2390	64	5.954840, 116.066929	5.932982, 116.088899
Sarawak	Betong	Batu Lintang	4336	64	1.535688, 110.343659	1.513709, 110.365639
Sarawak	Kuching	Padungan	2363	64	1.555490, 110.355554	1.533510, 110.377534
Sarawak	Kuching	Pending	2647	40	1.551161, 110.378983	1.529181, 110.400963
Sarawak	Miri	Pujut	2305	58	4.443012, 114.003883	4.421102, 114.025863
Sarawak	Kuching	Kota Sentosa	1898	26	1.473362, 110.329123	1.451382, 110.351103
Sarawak		Samariang	2652	32	1.637214, 110.329781	1.615234, 110.351761
Sarawak	Kuching	Satok	1446	43	1.553218, 110.334556	1.531238, 110.356536
Total Images				627		

5.2.2 Data Cleaning Steps

No explicit data cleaning was required, as the original data set was well structured and free of missing values, duplicates, or corrupted files. All images were intact and properly labeled, allowing for direct use in the pre-processing pipeline.

5.2.3 Feature Engineering and Transformation

To prepare the data for model training, several preprocessing and transformation steps were applied:

Resizing: All images were resized to a size of 224×224 pixels to ensure that they were compatible with convolutional neural network architectures such as ResNet-50 and VMamba.

Normalization: Pixel values were normalized to the range [0, 1] by dividing by 255, ensuring consistent input scales for the model.

Label Encoding: The region type (urban or rural) was encoded as binary values (urban = 1, rural = 0).

Train-Test Split: The dataset was split into 70% training, 15% validation, and 15% testing sets. The images were manually selected for each subset to ensure a balanced distribution between urban and rural classes.

5.3 Exploratory Data Analysis (EDA)

As part of the exploratory data analysis (EDA), we conducted a pixel count to understand the distribution of different classes in the images and performed an object count to analyze the frequency and density of various elements within the dataset. This helps assess data balance and guide preprocessing steps.

5.3.1 Pixel Count

Pixel count helps evaluate the distribution of the different land cover classes: Background, Buildings, Roads, Water, Barren, Forest, and Agriculture. Thus, it is possible to determine if this dataset truly reflects the characteristics of rural and urban environments. Rural regions are expected to exhibit a greater prevalence of Agricultural and Forest pixels, whereas urban regions are anticipated to demonstrate elevated quantities of buildings and roads. In our analysis of pixel counts, we incorporate a comparative boxplot. Moreover, both total and average pixel counts were analyzed to provide a complete understanding of the dataset distribution. We also provided a stacked bar chart to analyze the class distribution of all states.

5.3.1 (a) Boxplot of Pixel Count

The boxplot is shown in Figure 5.1. The boxplot effectively illustrates the central tendency, spread, and presence of outliers of each land-cover type in the dataset. The land-cover types include Background, Buildings, Road, Water, Barren, Forest, and Agriculture.

The different classes of land cover, such as water, forest and agriculture, within rural areas exhibit relatively high variations in their pixel counts compared to urban areas; the forest and agriculture classes show very high IQRs (476,648 and

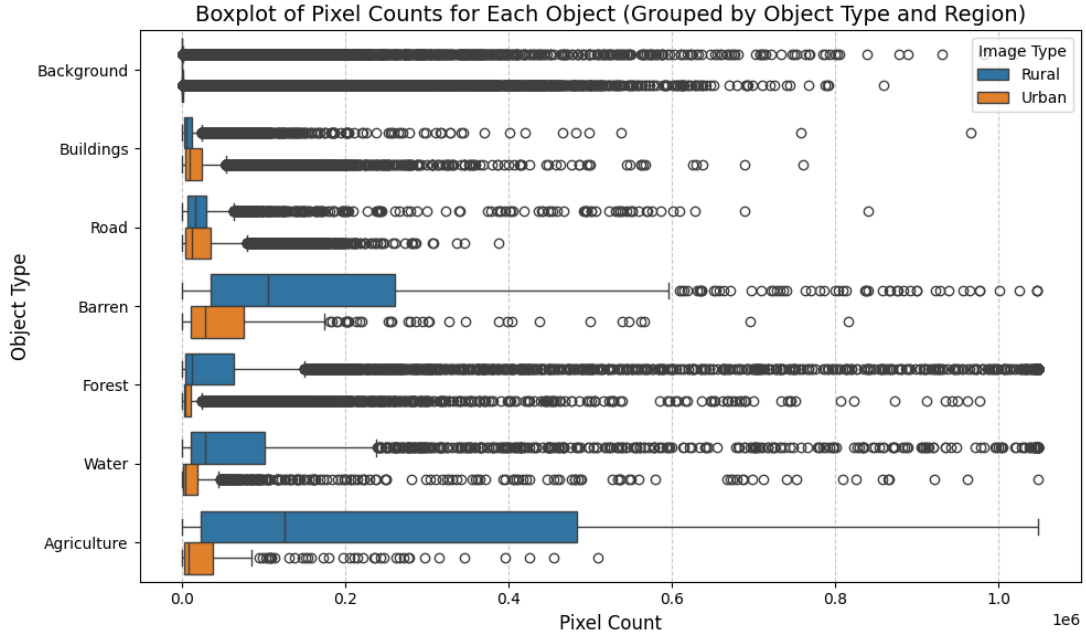


Figure 5.1: Pixel Count Boxplot of DeepEarthMY 2.0 Dataset

902,798 pixels, respectively), which emphasizes a very large variation within these types of land cover in the rural imagery. On the other hand, categories such as Buildings, Road, and Barren have smaller IQRs, indicating less variation and relatively stable pixel distributions. In terms of Water class, it is a part of the smaller IQRs in the rural imagery; however, it is larger than the Water class of urban imagery.

In the urban dataset, the Buildings class exhibits the largest IQR of 236,036.75 pixels, which indicates significant variation in the extent of building coverage across urban images. Forest shows large spreads with IQRs of 147915.25 pixels, while Road has a moderate range with an IQR of 80443.25 pixels. Thus, it highlights that road coverage varies considerably in urban environments compared to rural environments.

Interestingly, Water shows low pixel counts in most of the images with the lowest upper quartile value of 1,717.50 pixels. Furthermore, Agriculture and Barren appears to be absent with a quartile value and median of zero.

With regards to outlier detection, it becomes evident that almost all land-cover types of both rural and urban contain huge amounts of outliers; these are images with abnormally high pixel counts compared to the rest of the dataset. Rural outliers mainly consist of Buildings, Roads, Water, and Barren land-cover types; those outliers might represent cases of massive open fields, roads, urban structures, barren lands and large water bodies, which are rare but still valid in certain rural regions. While urban outliers are primarily made up of Water, Barren, Forest, and agricultural land-cover types, those outliers may constitute samples of gigantic water bodies, barren lands, forest, and agricultural land, which are rare but still found in a few urban regions.

5.3.1 (b) Barplot of Total Pixel Count

The bar chart in Figure 5.2 shows the total pixel count of each land-cover type. The total pixel count helps identify class imbalances across the entire dataset, which is crucial for evaluating the overall representation of each land cover class.

The rural dataset is dominated by Agriculture with a high pixel count of

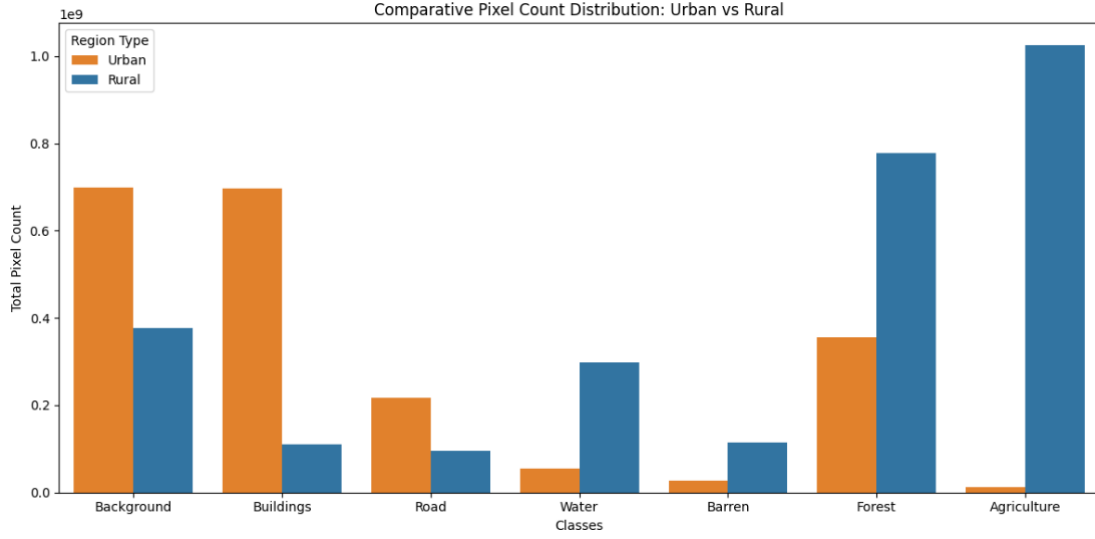


Figure 5.2: Pixel Count Barplot of DeepEarthMY 2.0 Dataset

1,024,404,489, followed by Forest with 778,036,624 pixels. These figures reflect the strong presence of open and green spaces in rural areas. The rural environment also has a moderate pixel count for Water (298,438,738 pixels) and Barren (114,654,614 pixels) compared to urban areas; this points out the heterogeneity of the rural environment with agriculture and natural aspects.

On the contrary, the urban dataset displays a completely different distribution pattern. Buildings are the most represented class with 696,167,353 pixels, which indicates the dense construction typical of urban environments. Despite this, the number of pixels assigned to Buildings in the urban dataset is only about seventy percent of the number of pixels assigned to Agriculture in the rural dataset. Moreover, the urban dataset includes roughly twice the number of Road pixels than the rural dataset does,

which means there is more road construction in urban places.

5.3.1 (c) Barplot of Average Pixel Count

As shown in Figure 5.3, the average pixel distribution varies significantly between rural and urban datasets. The barplot of average pixel counts truly reflects the differences in land cover characteristics.

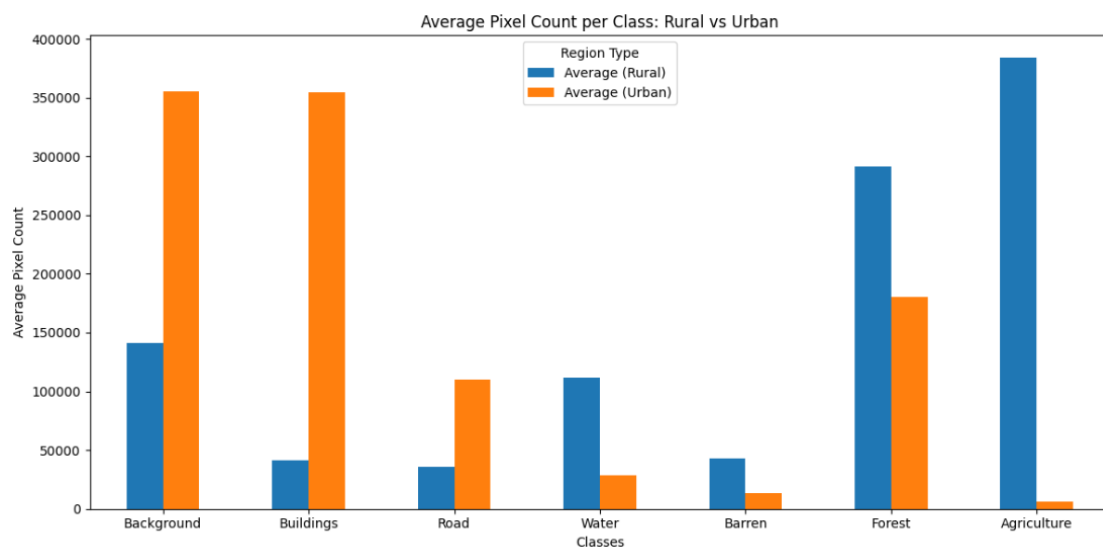


Figure 5.3: Average Pixel Count Barplot of DeepEarthMY 2.0 Dataset

To illustrate, the rural areas exhibit significantly higher mean pixel values for Water, Barren, Forest, and Agriculture. The high number of pixels in Agriculture mirrors the vast use of land for agriculture and farming, which is an integral part of rural economies. The increased count of pixels in the Forest would point to the strong existence of wooded areas, which is a characteristic of rural landscapes that still have large portions of their lands covered with forests. The high pixel count for water

class signifies considerable amounts in the lakes, rivers, and any other kinds of water bodies that are commonly found out in rural environments. Similarly, the high pixel count for Barren indicates the prevalence of uninhabited regions, for example: deserts or uncultivated land. Such areas of underdevelopment are usually characteristics of rural areas. The high average pixel values found in these classes represent the unique environmental and natural characteristics to native rural areas.

In contrast, urban images have a far higher share of Background, Buildings, and Road pixels. The high density of Background pixels denotes the densely built-up environments common in urban settings, where the landscape is usually dominated by man-made structures. Besides, the higher pixel count of Buildings point out that residential, commercial, and industrial buildings are well spread out in urban areas. Furthermore, the elevated pixel count associated with roads means that a great number of streets, highways, and transportation infrastructure in general are commonly found in metropolitan environments. Such high pixel counts in infrastructure-related land-cover classes reflect the processes of urbanization and human developments that are dominant in city landscapes, as opposed to the natural and agricultural focus of rural areas.

To summarise, the rural areas are mainly made up of agricultural fields, forests, and natural features, whereas the urban areas are characterized by a high

density of buildings, roads, and background pixels, all because of massive infrastructure. The image imbalance between rural and urban datasets significantly influences the distribution of the pixels. Still, image-quantity-based normalization steps have shown their effectiveness in reducing this issue. These analyses give more insight into the land cover dynamics of the dataset. They underscore the areas where we can improve by annotating additional images. Consequently, they will help us improve future model training and the accuracy of classification.

5.3.1 (d) Comparative Histogram of Land-Cover Types in Rural vs Urban Satellite

Images

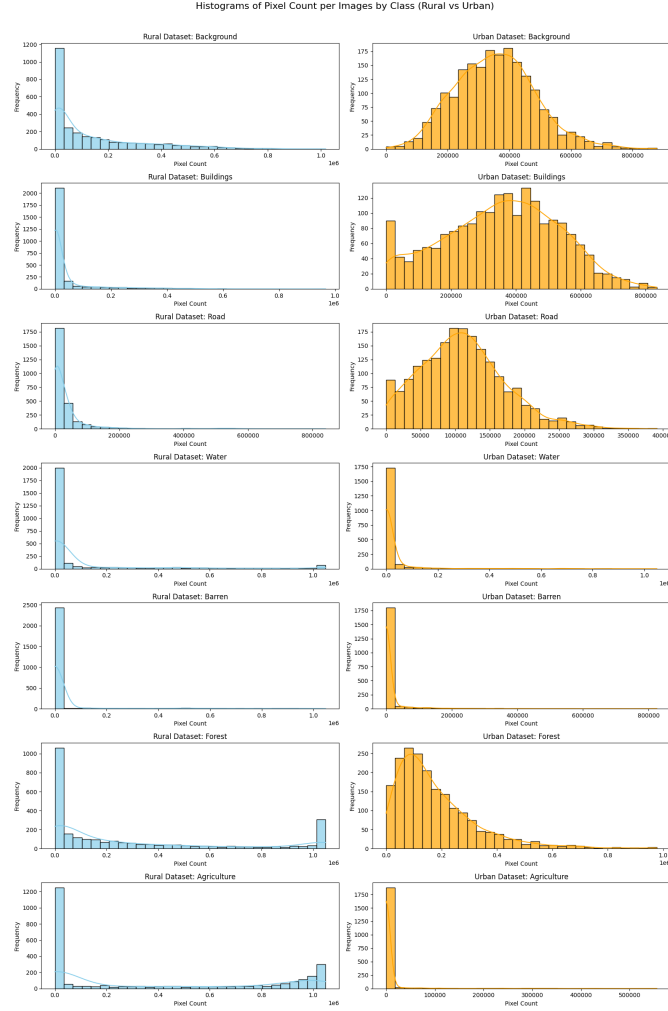


Figure 5.4: Comparative Histogram (Rural vs Urban)

The distribution of Buildings in rural areas is extremely right-skewed, indicating that most images have little to no Buildings pixel counts. For urban images, the Buildings class exhibits an approximately normal distribution, with outliers on the

lower end indicating that there are images with low building coverage.

The distribution of Road pixels in rural areas is right-skewed, showing that most images have minimal to no road coverage. The urban images also display a right skew, but it is not as extreme as rurals, indicating that most urban images have a low road pixel count, with fewer images having higher minimal road coverage.

Water pixels in rural and urban areas demonstrate a highly right skewed distribution, implying most rural images have little water coverage.

The Barren class in rural images shows a high right skew, suggesting that most areas have little barren land. For urban images, the Barren distribution is nearly identical, reflecting the little presence of barren land across different urban settings.

Forest pixels in rural images exhibit a strong right-skewed distribution with outliers on the high end. This suggests that most images have few forested areas, but some have high or extreme forest coverage. In urban areas, the Forest class shows an approximate right skew, reflecting that most urban images contain few forest pixels, with a few images including patches of green space.

Agriculture in rural images is heavily right-skewed with some outliers on the high end, indicating that some rural images contain extensive agricultural land, while a high number have little or no agriculture present. The urban agriculture distribution is right-skewed, demonstrating that agricultural land is generally scarce in urban areas.

5.3.1 (e) Stacked Bar Chart of Class Distribution by State

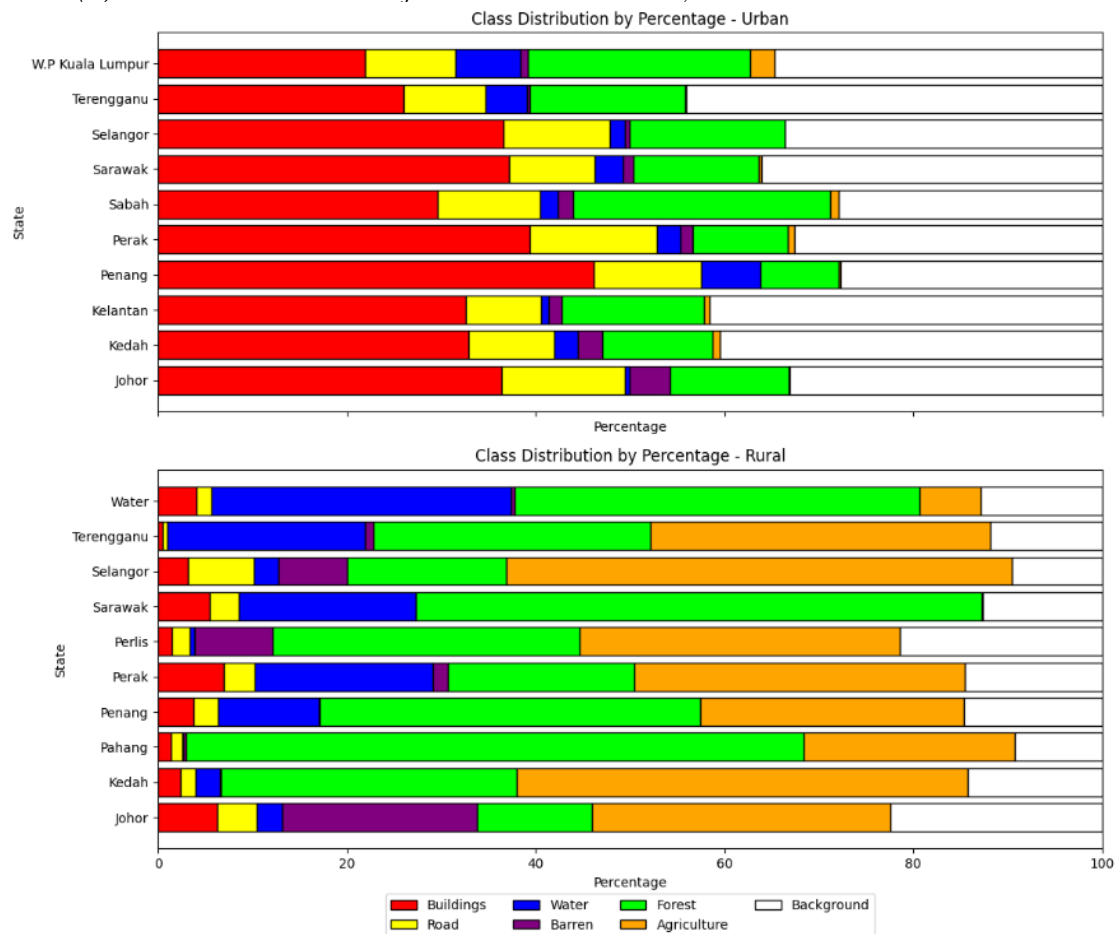


Figure 5.5: Stacked Bar Chart of Class Distribution by State

The stacked bar chart in Figure 5.5 shows the total pixel count of each land-cover type by state. The stacked bar chart is used to verify whether each state's

land cover distribution aligns with its expected classification as rural or urban.

The stacked bar chart illustrates that the rural-urban classifications of each state are correctly assigned. From the bar chart, we can see that Agriculture and Forest are more commonly found in each of the rural states, whereas all the urban states consist of pixels for Buildings and Roads. In addition, we can see that all the urban states have their fair share of forest pixels, but the number is not as significant as the rural forest count. We can conclude that the consistency found in the data indicates that all areas are labeled accurately and the land cover profile for each state is authentic.

5.3.2 *Object Count*

By offering information on the size and distribution of specific land-cover items, object count analysis enhances our comprehension of the distinctions between rural and urban areas and is a useful supplement to pixel count analysis. In addition, it assists in identifying outliers, possible biases, and inconsistent annotations in the dataset. The quality of the dataset for classification tasks is improved by combining the two analyses, which guarantees a more thorough representation of land-cover features.

5.3.2 (a) Comparative Boxplot of Pixel Counts for Each Object

Figure 5.2 below shows the comparative boxplot of pixel counts for each object.

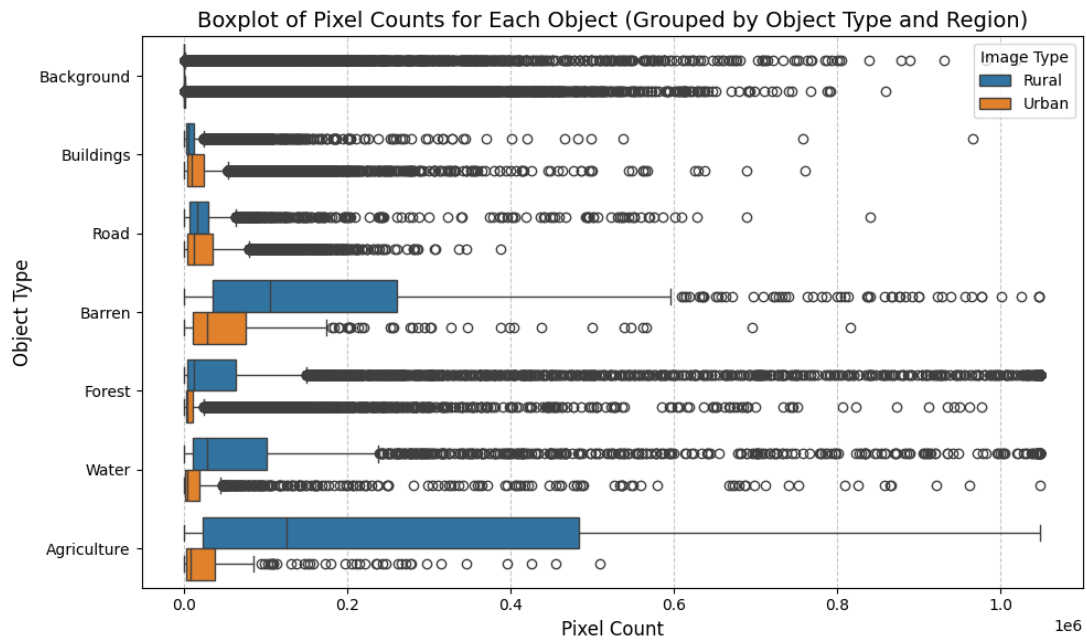


Figure 5.6: Comparative Boxplot of Pixel Counts for Each Object

Roads have relatively small pixel counts, concluding to limited Road size variation. Moreover, Barren exhibit high variability, which reflects diverse object sizes. Forest and water objects indicate clustered trees or small water bodies rather than large expanses. Agriculture has the highest spread, showing substantial variation in object sizes compared to urban areas.

In urban settings, Buildings are larger and more varied compared to rural

buildings. Roads in urban areas are also larger and more diverse. Conversely, barren, forest, water, and agricultural objects in urban areas have smaller pixel counts and less variation, with lower medians and narrower IQRs. Speaking about rural and urban, there are countless outliers for all the land cover types except Agriculture for rural. These outliers are unusually large objects. As it was mentioned before, the Background outliers are mainly uninhabited lands.

5.3.2 (b) Comparative Histogram of Average Pixel Count per Object

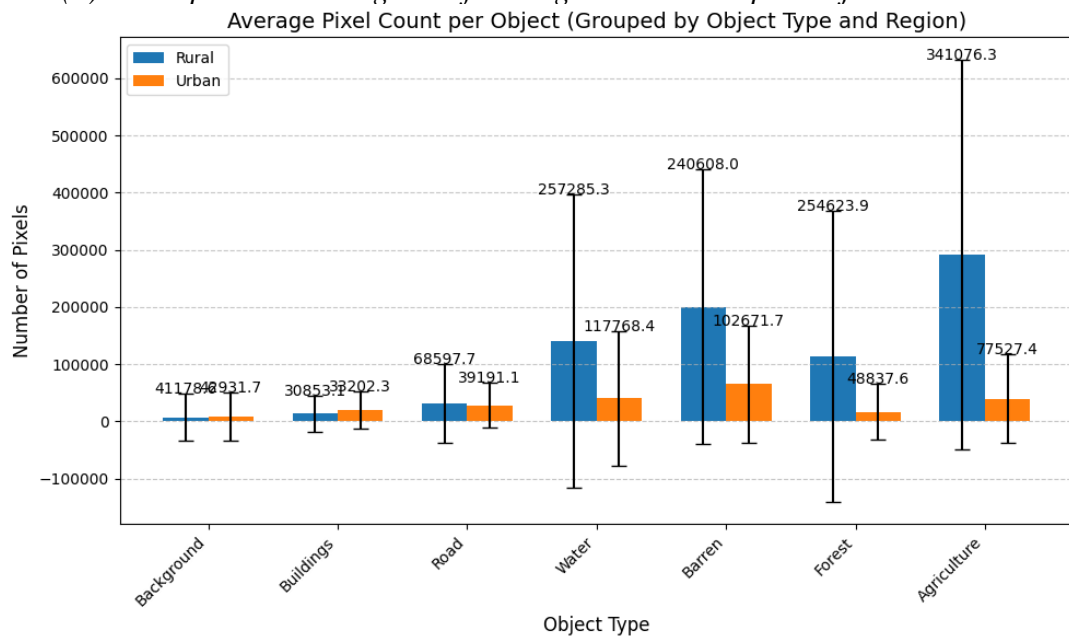


Figure 5.7: Comparative Histogram of Average Pixel Count per Object (Grouped by Object Type and Region)

In the rural region, Agriculture has the highest average pixel count per object (294,766.46). This shows that rural has bigger agricultural fields. Barren, Water, and Forest have high average pixel counts per object of 200,095.31, 140,305.01, and

114,746.78 respectively. This means the urban also contains large barren areas, water bodies, and forest regions. The average pixel counts per object of Road is 13,138.62. The barplot also shows that the Road value of rural is slightly larger than the Road value of urban. This might be explained by urban areas typically having a dense network of smaller roads, while rural areas may have fewer but wider roads. However, average pixel counts per object of Buildings is 31,628.72; this might be due to rural areas consisting of small buildings, which is logical.

In urban settings, the average pixel counts per object for Barren, Water, and Agriculture remain high. Although, they are smaller than the rural settings. This states that urban areas tend to have medium-sized agricultural land, water bodies, and barren lands compared to the expansive areas of rural environments. Moreover, urban forests have a much lower average pixel count per object than rural forests, indicating smaller and more fragmented green spaces. Urban roads also have a slightly lower average pixel count than rural roads, suggesting narrower or more segmented roads. Meanwhile, urban buildings have a higher average pixel count than rural buildings, reflecting larger or more clustered structures in cities.

The large standard deviations in both rural and urban settings are due to the considerable variety of object sizes between regions. Rural places include

different landscapes with wide agricultural fields, forests, and water bodies, whereas metropolitan areas contain a combination of dense skyscrapers, modest buildings, highways, and narrow streets, resulting in inconsistent pixel counts. Additionally, object fragmentation, such as roadways breaking into smaller portions in cities or varied widths in rural areas, adds to unpredictability. The dataset's use of several geographic regions and resolutions amplify these variances, representing real-world changes in land cover and infrastructure.

5.3.2 (c) Comparative Histogram of Land-Cover Types in Rural vs Urban Satellite

Images

Histograms of Pixel Count per Object by Class (Rural vs Urban)

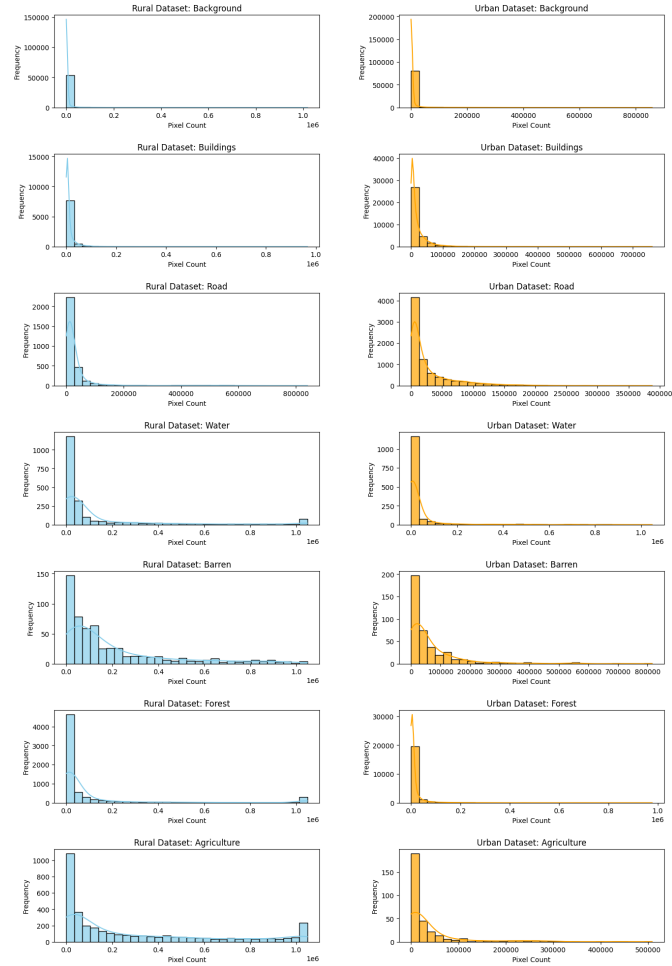


Figure 5.8: Comparative Histogram (Rural vs Urban)

The distribution of all the land-cover objects in rural and urban areas is extremely right-skewed. This is an indication that land-cover objects in the images are

spatially minor. However, the histograms of the classes of the 'Forest', 'Agriculture', and 'Water' in rural areas have outliers. These outliers states that there are few large objects in said categories, indicating the existence of extensive natural features, such as big patches of forests, crops fields, or lakes.

5.3.3 Conclusion of analysis

To summarise, object and pixel count studies provide complimentary information about rural and urban land cover disparities. Rural areas have larger agricultural fields, woodlands, and water bodies, whereas urban areas have more diverse but smaller buildings and roads. Boxplots and histograms demonstrate considerable differences, outliers, and annotation discrepancies, with rural parts having higher pixel spreads in Agriculture and Forest, whilst urban areas have more diverse building and road sizes. Object count reveals that rural areas have larger but fewer objects, whereas urban environments have more fragmented land-cover elements. These findings improve dataset quality by providing a more complete representation of land-cover features for categorization purposes.

5.4 Model Implementation

5.4.1 Algorithms/Models Used

Several deep learning models were employed to evaluate the performance of land-cover classification from satellite images. These included:

- **AlexNet** – one of the earliest deep convolutional neural networks (CNNs) that helped pioneer deep learning in computer vision. It served as a baseline model for comparison, utilizing stacked convolutional and fully connected layers.
- **EfficientNet** – a family of CNNs that scale depth, width, and resolution efficiently. It balances accuracy and computational cost by optimizing the model architecture through compound scaling.
- **EfficientNetV2**: An improved version of EfficientNet with faster training, better accuracy, and more efficient use of parameters. It introduces progressive learning, optimized scaling, and Fused-MBConv blocks for improved performance.
- **ResNet-50** – a deep residual network with 50 layers, known for its residual connections that allow the model to train effectively even at great depth by mitigating the vanishing gradient problem.

- **ViT (Vision Transformer)** – a transformer-based model for image classification that treats images as sequences of patches, allowing it to learn long-range dependencies using self-attention mechanisms.
- **VMamba** – a vision model based on the Mamba architecture, using state space models to efficiently capture spatial and long-range dependencies. It achieves linear complexity and strong performance on high-resolution image tasks

These models were selected to compare the performance of traditional CNN-based architectures and newer transformer-based and hybrid models for land-cover classification.

5.4.2 Training and Testing Procedures

The dataset was split into training (70%), validation (15%), and testing (15%) sets. This is done to:

- Maintain spatial independence between training, validation, and test sets.
- Avoid placing images from the same district into both training and validation/test sets.
- Ensure a balanced representation of urban and rural images across all sets.

Table 5.5: Final Data Split After Adjusting for Spatial Independence

Domain	Train	Val	Test
Urban	1344	312	309
Rural	1878	389	402
Total	3222	701	711

- Follow a 70:15:15 ratio for data splitting:
 - Training: 70%
 - Validation: 15%
 - Testing: 15%
- Make sure both urban and rural images are included in the validation and test sets to properly evaluate generalization across domains.

By following a 70:15:15 split ratio to divide our dataset into training, validation, and test sets. Based on this ratio, the ideal distribution is:

Table 5.6: Distribution and Split of Urban Annotated Images in DeepEarthMy 2.0 Dataset

Domain	State	District	Constituency / State Assembly	Population Density	#Images	Train	Val	Test
Urban	Johor	Johor Bahru	Kota Iskandar	1210	64	✓		
		Johor Bahru	Larkin	4963	63		✓	
	Kelantan	Kota Bahru	Kota Lama	4075	64	✓		
	W.P Kuala Lumpur	W.P Kuala Lumpur	Bukit Bintang	5739	64			✓
	W.P Putrajaya	W.P Putrajaya	W.P Putrajaya	2229	125	✓		
	Penang	Timur Laut	Komtar	3913	64			✓
	Sabah	Kota Kinabalu	N.21 Luyang	5440	64	✓		
		Kota Kinabalu	Api-Api	5251	57	✓		
		Sandakan	Elopura	5339	54			✓
		Sandakan	Sungai Sibuga	2045	63			✓
		Tawau	Kukusan	3737	62	✓		
		Putatan	Petagas	2390	64	✓		
	Selangor	Gombak	Bukit Antarabangsa	8820	64			✓
		Petaling	Seri Setia	5735	64	✓		
		Petaling	Kampung Tunku	7842	64	✓		
		Klang	Pandamaran	5663	63		✓	
		Petaling	Batu Tiga	5619	63	✓		
		Petaling	Kota Anggerik	2567	64	✓		
		Petaling	Subang Jaya	8320	64	✓		
		Kedah	Kota Setar	Kota Darul Aman	5773	64		✓
	Kedah	Kuala Muda	Bakar Arang	2858	64	✓		
		Perak	Kinta	Menglembu	2168	64	✓	
	Perak	Kinta	Buntong	4613	64	✓		
		Sarawak	Sibu	Pelawan	3498	64		✓
	Betong		Batu Lintang	4336	64	✓		
	Kuching		Padungan	2363	64	✓		
	Kuching		Pending	2647	40	✓		
	Kuching		Kota Sentosa	1898	26	✓		
	Kuching		Samariang	2652	32	✓		
	Kuching		Satok	1446	43	✓		
	Miri		Pujut	2305	58		✓	
	Terengganu	Dungun	Sura	1942	64	✓		
Urban Total					1965	1344	312	309

Table 5.7: Distribution and Split of Rural Annotated Images in DeepEarthMy 2.0 Dataset

Domain	State	District	Constituency / State Assembly	Population Density	#Images	Train	Val	Test
Rural	Johor	Muar	Sungai Balang	199	64			✓
		Johor Bahru	Tiram	592	128	✓		
	Kedah	Langkawi	Ayer Hangat	150	64	✓		
		Langkawi	Kuah	290	64	✓		
		Yan	Sungai Limau	287	64	✓		
		Pendang	Tokai	270	63	✓		
		Kubang Pasu	Bukit Kayu Hitam	167	64		✓	
	Penang	Kota Setar	Bukit Pinang	511	63			✓
		Barat Daya	Pulau Betong	621	64		✓	
	Pahang	Cameron Highlands	Tanah Rata	57	127	✓		
		Cameron Highlands	Jelai	12	64		✓	
	Perak	Kerian	Titi Serong	609	190	✓		
		Hilir Perak	Teluk Intan	974	128	✓		
		Manjung	Pantai Remis	161	64		✓	
		Batang Padang	Sungkai	118	64	✓		
		Kampar	Malim Nawar	210	118	✓		
		Muallim	Slim	55	63			✓
	Perlis	Perlis	Chuping	162	62			✓
		Perlis	Pauh	777	63	✓		
	Selangor	Sabak Bernam	Sungai Air Tawar	98	64		✓	
		Sepang	Sungai Pelek	284	126	✓		
		Klang	Meru	1438	20			✓
		Kuala Selangor	Bukit Melawati	295	13		✓	
		Sabak Bernam	Sungai Panjang	101	192	✓		
		Sabak Bernam	Sabak	190	64			✓
		Kuala Selangor	Permatang	57	62	✓		
		Sabak Bernam	Sekinchau	79	92	✓		
		Sarawak	Bintulu	50	12		✓	
	Terengganu	Kuala Nerus	Batu Rakit	270	21			✓
		Setiu	Permaisuri	53	127	✓		
	Water	-	-	-	295	206	44	45
Rural Total					2669	1878	389	402
Grand Total					4634	3222	701	711

All models were trained using the training set and tuned on the validation set. Final performance was evaluated on the test set. Training was conducted using mini-batch gradient descent with a batch size of 16 and early stopping based on validation loss. VMamba is trained for 300 epoch while to other models were trained on 100 epoch.

5.4.3 *Hyperparameter Tuning*

The following parameter settings were used for each model implementation. These parameters are based on commonly adopted configurations from prior literature or official repositories. No additional hyperparameter tuning or fine-tuning was conducted:

- **AlexNet:** Batch size = 16, epochs = 100, learning rate = 1e-3, optimizer = {opt}, dropout rate = {0.5, 0.5}, metrics = accuracy.
- **EfficientNet (B5, B7):** Batch size = 16, epochs = 100, Initial learning rate = 1e-3, learning rate scheduler = ExponentialDecay (decay rate = 0.9 every 10,000 steps), optimizer = {Adam}, dropout rate = {0.5, 0.5}, metrics = accuracy.
- **EfficientNetV2 (B0, B1, B2):** Batch size = 16, epochs = 100, Initial learning rate = 1e-3, learning rate scheduler = ExponentialDecay (decay rate = 0.9 every 10,000 steps), optimizer = {Adam}, dropout rate = {0.5, 0.5}, metrics = accuracy.

- **EfficientNetV2-B3**: Batch size = 16, epochs = 100, learning rate = $1e-3$, optimizer = {Adam}, dropout rate = {0.5, 0.5}, metrics = accuracy.
- **ResNet-50**: Learning rate = {0.01, 0.001}, optimizer = {Adam, SGD}, dropout rate = {0.3, 0.5}.
- **Vision Transformer (ViT)**: Batch size = {32, 64}, epochs = 100, learning rate = { $3e-4$, $1e-4$ }, optimizer = AdamW, weight decay = {0.01, 0.03}, attention heads = 12, embedding dimension = 768.
- **VMamba (Tiny, Small, Base)**: Batch size = 16, epochs = 300, base learning rate = $1.5625e-5$, optimizer = AdamW, weight decay = 0.05, learning rate scheduler = Cosine, warmup epochs = 20, drop path rate = 0.5, dropout rate = 0.0, label smoothing = 0.1.

5.4.4 *Performance Metrics Used*

The following performance metrics were used to evaluate and compare models:

- **Accuracy**: The proportion of correctly classified samples over the total number of samples.
- **Loss**: A scalar value that reflects how well the model's predicted probabilities

match the actual labels. Lower loss indicates better model confidence and convergence.

- **Precision:** The ratio of true positives to all predicted positives, indicating the relevance of positive predictions.
- **Recall:** The ratio of true positives to all actual positives, indicating how well the model captures relevant cases.
- **F1-Score:** The harmonic mean of precision and recall, especially useful for imbalanced datasets.
- **Confusion Matrix:** A matrix representation of prediction results showing true vs predicted labels.

CHAPTER 6

EVALUATION OF FINDINGS

6.1 Overall Model Comparison

Table 6.1 presents a comprehensive summary of how well each model performs in land-cover classification across multiple evaluation metrics. The experiment is conducted on DeepEarthMY version 2.0. The best-performing epoch for each model was selected based on the highest validation accuracy.

Table 6.1: Comprehensive Model Performance Comparison

Model	Accuracy	Loss	Precision	Recall	F1-score
vMambaV2-Small	0.8453	0.3659	0.8043	0.8511	0.8270
ResNet50	0.8467	0.8403	0.7463	0.9806	0.8476
EfficientNetV2-B2	0.9100	0.6098	0.8393	0.9806	0.9045
AlexNet	0.7792	0.4691	0.7021	0.8544	0.7708
ViT	0.7876	0.4637	0.7257	0.8220	0.7709

- **vMambaV2-Small:**

vMambaV2-Small shows consistent and reliable performance with accuracy of 84.53%, precision of 0.8043, and recall of 0.8511. It also achieves the lowest loss of 0.3659, indicating that the model is not only making accurate predictions but doing so with higher confidence. The F1-score of 0.8270 reflects this balance

between precision and recall. Its efficient learning and compact structure make it a promising option when computational resources are limited.

- **ResNet50:**

ResNet50 performs competitively in terms of accuracy 84.67% and F1-score of 0.8476, but shows a high loss of 0.8403, the worst among all models, which suggest overfitting. Its extremely high recall of 0.9806 and lower precision of 0.7463 suggest the model predicts almost all actual positives correctly, but at the cost of a larger number of false positives. This precision-recall imbalance may reflect over-sensitivity, especially in noisy or ambiguous samples. The high loss further suggests unstable or low-confidence predictions despite its solid classification metrics.

- **EfficientNetV2-B2:**

EfficientNetV2-B2 with learning rate decay demonstrates the **best overall performance**, achieving the highest accuracy of 91.00% and F1-score of 0.9045 among all models. With a high recall of 0.9806, it effectively captures true positives, while maintaining strong precision of 0.8393. This indicates balanced performance with minimal false positives or false negatives. The moderate loss of 0.6098 suggests good convergence, though there may be room for improved confidence calibration. Overall, this model exhibits excellent generalization and is the most suitable candidate for deployment.

- **AlexNet:**

AlexNet is the weakest model overall, with the lowest accuracy of 77.92% and F1-score of 0.7708, although its recall is respectable at 0.8544. This suggests that the model is identifying most true positives but suffers from low precision of 0.7021, implying a higher number of false positives. The loss of 0.4691 is relatively low, showing that the predictions are made with some confidence, but the overall architecture is likely too outdated to compete with more modern alternatives.

- **ViT (Vision Transformer):**

ViT achieves modest performance with an accuracy of 78.76% and an F1-score of 0.7709. While it maintains relatively good recall of 0.8220, its precision is lower at 0.7257, which may indicate a tendency toward false positives. The loss value of 0.4637 is lower than most CNN models, showing decent model calibration. However, ViT appears to underperform relative to more optimized architectures like EfficientNetV2, likely due to its data-hungry nature and lack of convolutional inductive biases.

6.2 *EfficientNet and EfficientNetV2 Variant Comparison*

6.2.1 *Evaluation Metrics*

After overall model comparison, we wanted to compare the classification performance across different versions of the EfficientNet family, multiple variants were evaluated, including EfficientNetV2 (B0, B1, B2, B3) and EfficientNet (B5, B7). The purpose of this comparison is to determine which version offers the best predictive accuracy and overall effectiveness for land-cover classification.

Table 6.2: Classification Performance of EfficientNet Variants

Model	Accuracy	Loss	Precision	Recall	F1-score
EfficientNetV2-B0	0.8861	0.3008	0.8016	0.9806	0.8821
EfficientNetV2-B1	0.8650	0.4623	0.7766	0.9676	0.8617
EfficientNetV2-B2	0.9100	0.6098	0.8393	0.9806	0.9045
EfficientNetV2-B3	0.9001	0.3455	0.8324	0.9644	0.8936
EfficientNet-B5	0.8579	0.6007	0.7640	0.9741	0.8563
EfficientNet-B7	0.9058	0.5985	0.8306	0.9838	0.9007

From Table 6.2, it can be observed that EfficientNetV2-B2 has the best accuracy and precision of 0.9100 and 0.8393, respectively. This shows that EfficientNetV2-B2 is best at correctly classifying true positives and decreasing false positives. This level of performance makes this variant very reliable when overall classification is required.

However, it should be noted that EfficientNetV2-B2 does outperform in every other metric. EfficientNetV2-B0 has the least loss of 0.3008, which means that it will make more stable and confident predictions, despite lowering accuracy and F1-score by a small margin. Besides, EfficientNet-B7 has the best recall of 0.9838; therefore, it shows the best ability to recognize almost all the instances of true-positives. Such an aspect makes B7 especially useful in cases where maintaining a low proportion of false negatives is key.

In terms of F1 score, EfficientNetV2-B2 has reached the highest value, followed by EfficientNet-B7, and EfficientNetV2-B3. These results confirm that despite other models having high recall or loss scores, EfficientNetV2-B2 is able to demonstrate the most balanced and even performance across the metric.

All in all, the EfficientNetV2-B2 can be deemed as the most precise and accurate model, as its F1-score is an indicator of its reliance both on avoiding false positives and false negatives. However, variants, like B0 or B7 may have their own unique benefits, depending on particular use-case requirements, for instance: favoring model confidence, sensitivity, or computation speed.

6.2.2 Training and Validation Performance

The figures below illustrate the training and validation accuracy and loss for the EfficientNet variants. These plots provide further insight into the learning dynamics of each model, highlighting convergence speed, potential overfitting, and stability.

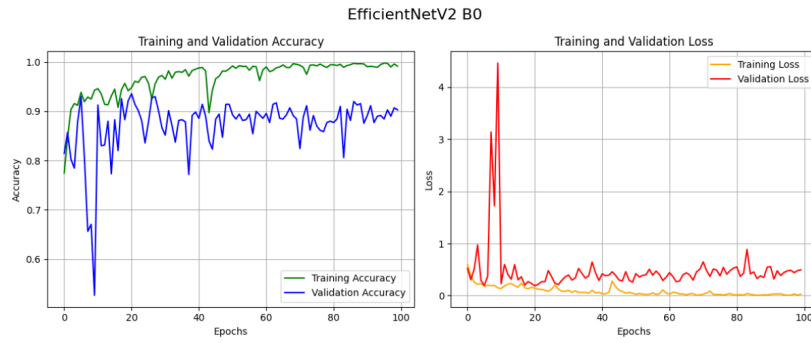


Figure 6.1: Training and validation of EfficientNetV2-B0 over epochs.

In EfficientNetV2-B0, the training curves show that the model performs well in the training set with accuracies nearing 1.0 and low loss values. However, the validation accuracy sharply dips to 0.2 around Epoch 10, and it fluctuates across epochs, mostly ranging from 0.85 to 0.92. Additionally, the validation loss has several sharp spikes between epochs 2–15. After that, the validation loss begins to stabilize, but it shows an increasing trend till the end. This might be due to slight overfitting.

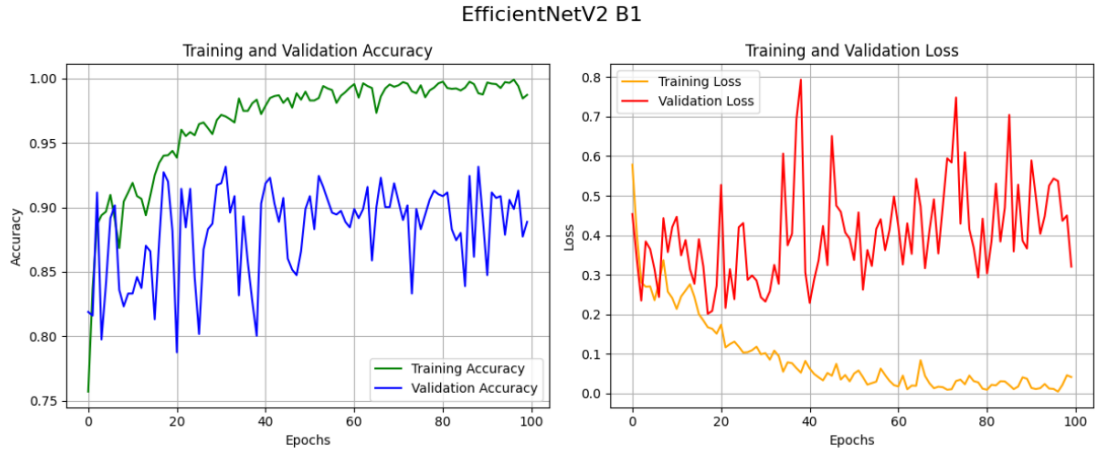


Figure 6.2: Training and validation of EfficientNetV2-B1 over epochs.

EfficientNetV2-B1 is learning slower than V2-B0, but still reaches accuracies near 1.0 and low loss values. On the other hand, the validation accuracy constantly fluctuates around 0.80 to 0.93. Moreover, the validation loss is keeps fluctuating with a magnitude of around 0.3, and it also shows an increasing trend, although it has no significant spikes. Therefore, this model shows a more stable behavior compared to V2-B0.

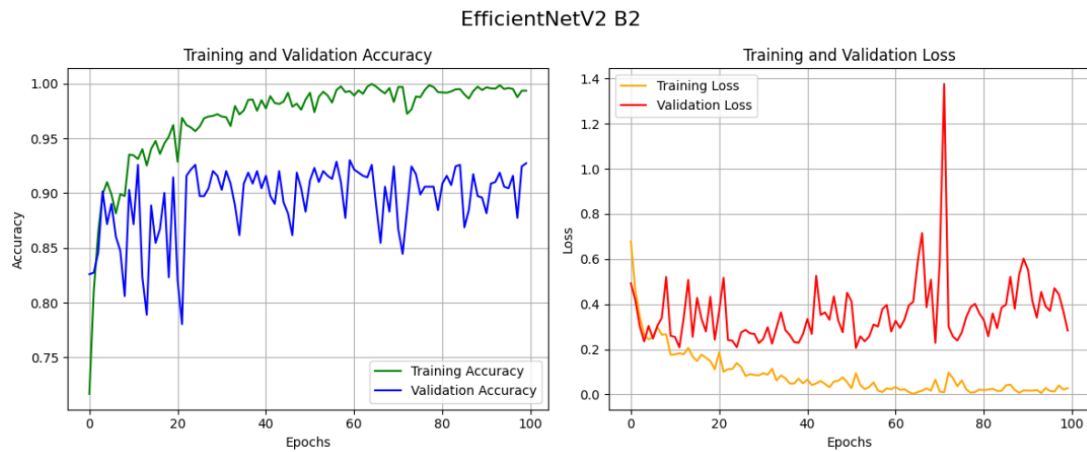


Figure 6.3: Training and validation of EfficientNetV2-B2 over epochs.

In EfficientNetV2-B2, the model performs greatly in the training set with accuracies nearing 1.0 and low loss values. However, the validation accuracy is high, but it fluctuates across epochs, mostly ranging from 0.85 to 0.92. Additionally, the validation loss is relatively stable with fluctuations ranging from 0.25 to 0.5 and a sharp spike of 1.4 loss around epoch 70. Even with the consistencies in the validation loss, the model achieved the best overall performance.

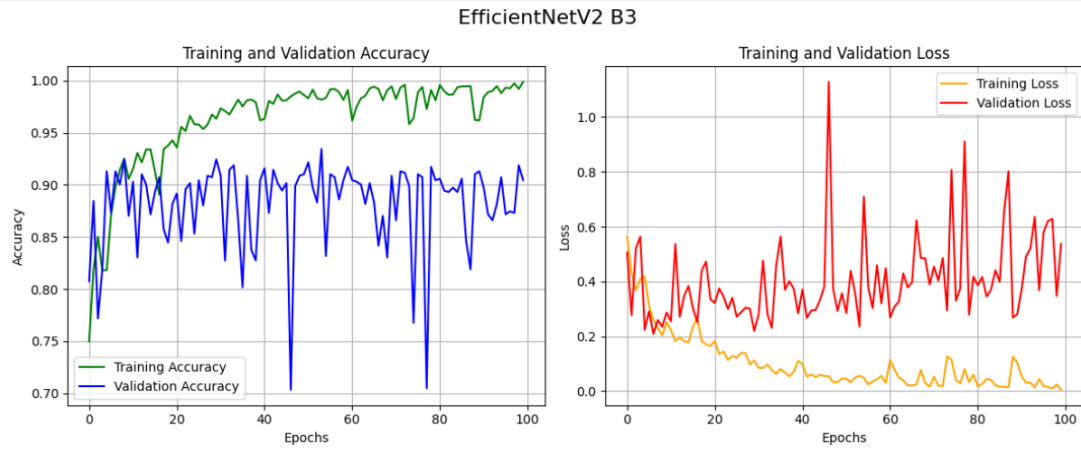


Figure 6.4: Training and validation of EfficientNetV2-B3 over epochs.

For EfficientNetV2-B3, the training curves also display accuracies nearing 1.0 and low loss values, but the accuracy dropped once in the early stage. In terms of validation, the accuracy fluctuation is more pronounced compared to the validation accuracy of EfficientNetV2-B2, with several medium drops across epochs. In addition, the validation loss is also more pronounced compared to the validation loss of EfficientNetV2-B2. This is due to fluctuations that vary within a range of approximately 0.2 and a sharp spike of approximately 1.1 at around epoch 45. It can be concluded that this model has a less stable generalization than B2.

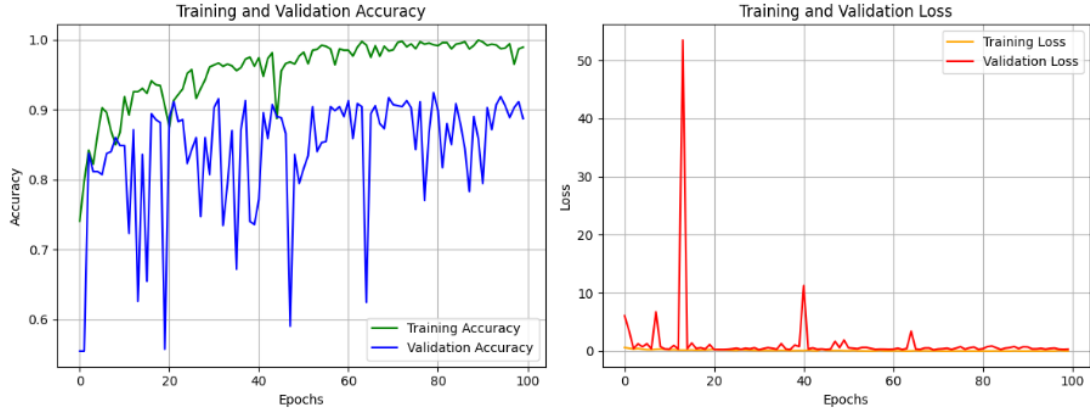


Figure 6.5: Training and validation of EfficientNet-B5 over epochs.

For EfficientNet-B5, the training curves also display accuracies nearing 1.0 and low loss values. Furthermore, the validation accuracy fluctuates at a magnitude of 0.3 across epochs. In addition, the validation loss shows an extremely high spike of 53 around epoch 15 and several spikes around epochs 8, 40, and 65, respectively. Such anomalies indicate that even when applying learning rate decay, the model is still overfitting and cannot generalize properly. This could be explained by the model's sensitivity to noise in the validation set or the model's high capacity relative to the dataset size. Therefore, EfficientNet-B5 might not be the most appropriate model for our task.

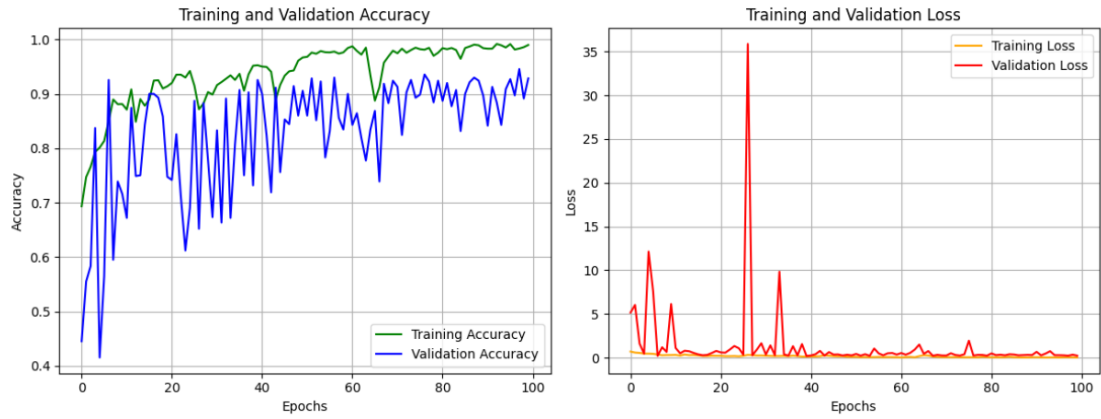


Figure 6.6: Training and validation of EfficientNet-B7 over epochs.

For EfficientNet-B7, the training accuracies also increase near 1.0 with slight fluctuations, and its training loss is normal. In addition, the validation accuracy fluctuates very noticeably at the beginning, but fluctuations decrease as the epochs increase. In terms of validation loss, the fluctuation is similar to EfficientNet-B5, showing an extremely high spike of 35 around epoch 27 and several spikes around epochs 5, 10, and 35, respectively. In conclusion, the model is still overfitting and cannot generalise properly, like EfficientNet-B5. Therefore, EfficientNet-B7 might also not be an appropriate model for our task.

6.3 VMamba Variant Comparison

6.3.1 Evaluation Metrics

Since VMamba was selected as the final architecture in the research methodology, a detailed comparison of its variants is essential to identify the most suitable configuration within the family.

Table 6.3: Classification Performance of vMamba Variants

Model	Accuracy	Loss	Precision	Recall	F1-score
VMambaV2-Tiny	0.8158	0.4227	0.8049	0.5340	0.6420
VMambaV2-Small	0.8453	0.3659	0.8043	0.8511	0.8270
VMambaV2-Base	0.8158	0.4314	0.8296	0.7249	0.7737

As indicated in Table 6.3, VMambaV2-Small shows the best overall accuracy level of 0.8453, lowest level of loss, and highest F1-score of 0.8270. Thus, this model is balanced and provides stability in predicting both positive and negative samples. VMambaV2-Small also excels in detecting positive instances with few false negative rates as shown with a recall of 0.8511.

Other than that, VMambaV2-Base is the most precise model, with a precision of 0.8296. This means that the model has more chance of being right compared to the other two models. However, this increased precision caused a recall of 0.7249, which leads to a decreased F1-score of 0.7737 and a less balanced performance in general.

This can prove beneficial in situations where the cost of a false positive exceeds false negative.

VMambaV2-Tiny has the poorest performance towards most of the metrics. Its precision of 0.8049 is quite satisfying, but its recall of 0.5340 is the poorest, resulting in the least F1-score of 0.6420. It also has the highest corresponding loss value of 0.4227. Although it can still make confident predictions, VMambaV2-Tiny fails to identify many positive cases numerous times. This suggests that it would not work well in general classification tasks.

In conclusion, it is necessary to note that VMambaV2-Small presents the most balanced and accurate model with high performance in all evaluation metrics and excellent generalization ability. The practical value of vMambaV2-Base can be higher when precision is of more importance. Moreover, MambaV2-Tiny is lightweight but it may not be suitable for task that need balanced classification performance.

6.3.2 Training and Validation Performance

The figures give a detailed description of the accuracy and loss training and validation plots of each of the VMamba variants. Plots can show important information regarding the learning dynamics of the models, such as the speed of convergence,

possible overfitting, and stability.

It is worthy to note that the training accuracy is lower than the validation accuracy. This counterintuitive behavior is primarily due to strong data augmentation techniques—such as Mixup, CutMix, and AutoAugment, which deliberately make the training data more challenging by distorting images and blending labels. Additionally, regularization methods like DropPath and label smoothing further penalize overconfident predictions during training. Since these techniques are typically disabled during validation, the model performs better on the cleaner, unaugmented data, resulting in higher validation accuracy than training accuracy. It is also best to note that this phenomenon similarly explains why the validation loss often appears lower and smoother than the training loss. The training loss reflects the impact of augmentations, label noise, and regularization, whereas the validation loss is computed on clean data in evaluation mode.

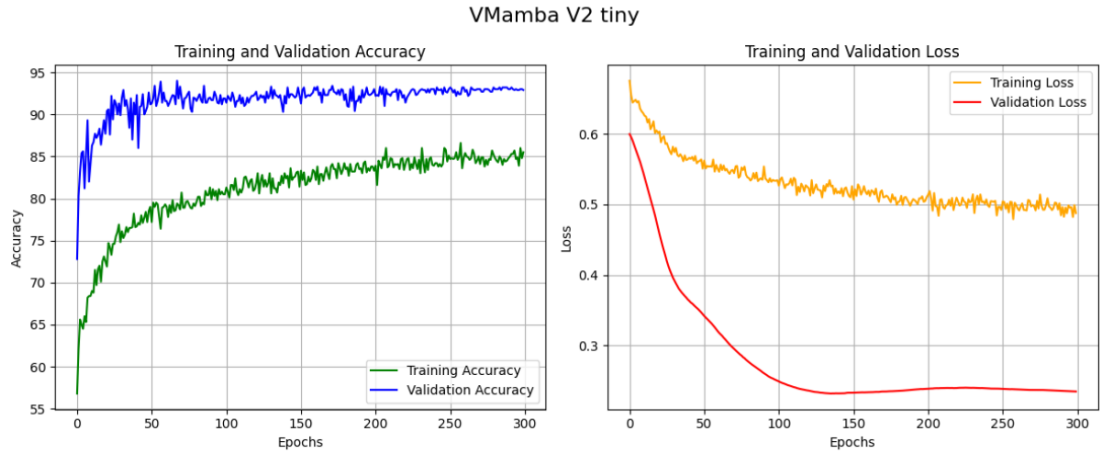


Figure 6.7: Training and validation of VMambaV2-Tiny over epochs.

In VMambaV2-Tiny, the validation curve shows that the model performs well, with accuracies reaching 94%. We can also see that the validation accuracy has fluctuations at magnitudes of 4. Moreover, the validation loss curve remains smooth and steadily decreases from epoch 0 to around epoch 125, after which it begins to increase slightly, indicating a potential onset of overfitting.

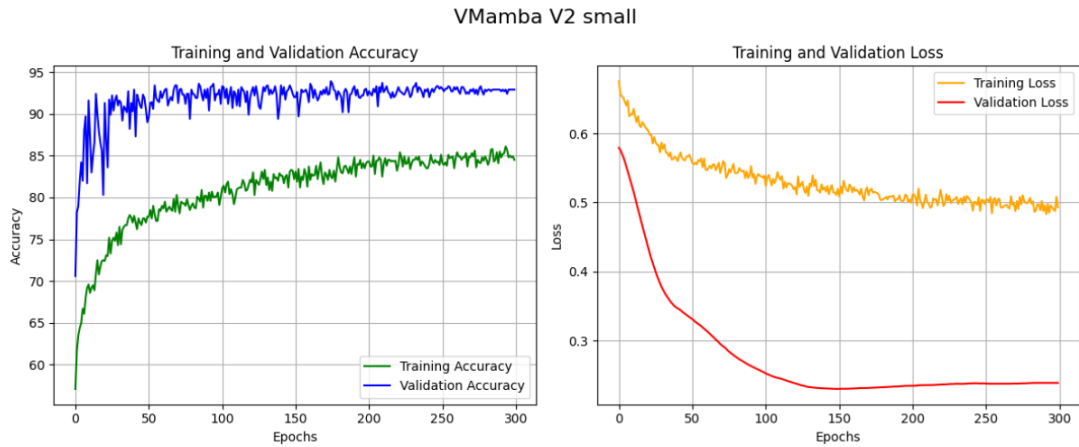


Figure 6.8: Training and validation of VMambaV2-Small over epochs.

In VMambaV2-Small, the validation curve shows that the model also performs well, with accuracies reaching 94%. However, we can see that the validation accuracy has slightly larger fluctuations. Moreover, the validation loss curve show behaviors similar to Tiny, where the loss decreases from epoch 0 to around epoch 125, after which it begins to increase slightly. It is important to note that this model is slightly more accurate in comparison to VMambaV2-Tiny which means that the added capacity of the Small version provides a better performance, although it is associated with risk of overfitting.

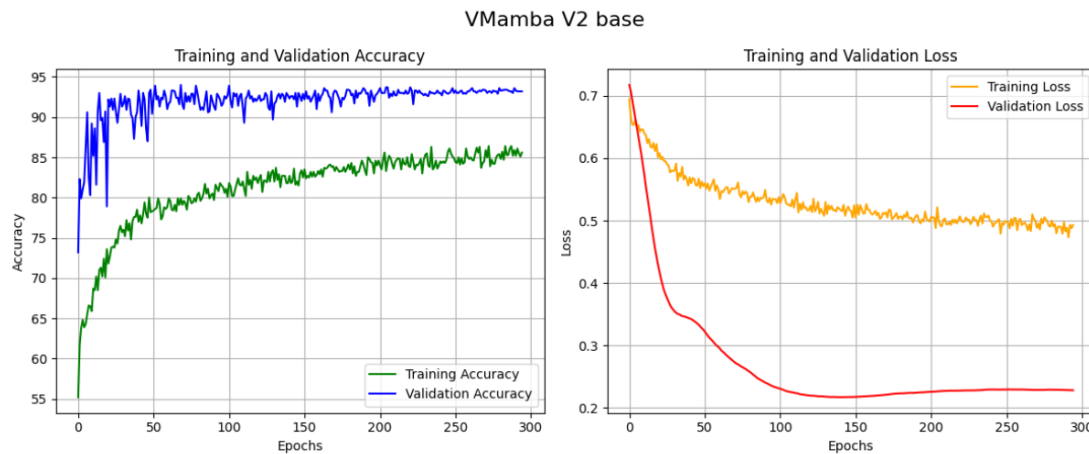


Figure 6.9: Training and validation of VMambaV2-Base over epochs.

In VMambaV2-Base, the validation curve shows that the model also performs well, with accuracies reaching 94%. However, we can see that the validation accuracy has similar fluctuations to the Small version. Moreover, the validation loss curve show that the loss decreases from epoch 0 to around epoch 125, with a slight bump

around epoch 40. After that, it begins to increase slightly similar to the other versions. Despite having a larger number of parameters and higher capacity, the Base model's performance is very similar to Tiny's. This implies that increased complexity does not guarantee better performance on this dataset. This might be because the complexity of land-cover classification or dataset size limitations.

6.3.3 *DeepEarthMY Version 1.0 Performance*

After conducting extensive experiments on DeepEarthMY Version 2.0 (V2), the best-performing model was selected based on its overall classification performance, which is EfficientNetV2-B2. To assess how the dataset version impacts model accuracy, we further evaluated the best model on both versions of the dataset. The datasets will be randomly split into training, validation, and test sets. This comparison aims to highlight the effect of the additional annotated images in version 2.0 and to assess the effectiveness of the manual split used in version 2.0.

The results of this experiment are shown below:

Table 6.4: Comparison of Model Performance on Different Dataset Versions and Splits

Dataset Version	Split Method	Split Ratio	Accuracy	Loss	Precision	Recall	F1 Score
V1	Random	80,10,10	0.9526	0.1703	0.91457	0.9104	0.9278
V2	Random	75,15,15	0.9526	0.2499	0.9396	0.9492	0.9444
V2	Manual	75,15,15	0.9100	0.6098	0.8393	0.9806	0.9045

Table 6.4 presents a comparison of model performance across different dataset

versions and splitting methods. Notably, the manually split V2 dataset yields the lowest accuracy and highest loss among the three configurations, despite achieving a high recall. This suggests that while the model is able to correctly identify most positive instances, it suffers from reduced overall performance, likely due to class imbalance or challenging examples concentrated in the test set. In contrast, both random splits on V1 and V2 produced higher accuracies of 95.26%, with V2 random showing stronger precision of 0.9396 and F1 score of 0.9444, while V1 random had slightly lower loss of 0.1703 and more balanced metrics overall. These results indicate that the manual split in V2 may not be optimal, possibly introducing unintended bias or difficulty that the model struggles to generalize across. Thus, the experiment highlights that although manual splitting is often intended to improve robustness, in this case, the manually split V2 may not be suitable and requires reconsideration or refinement for future experiments.

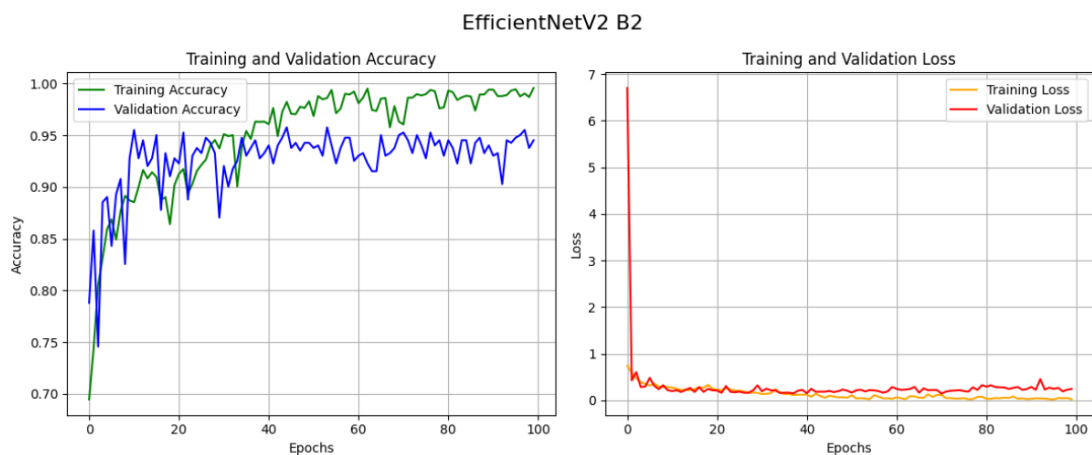


Figure 6.10: Training and validation of EfficientNetV2-B2 on DeepEarthMY Version 1.0 over epochs.

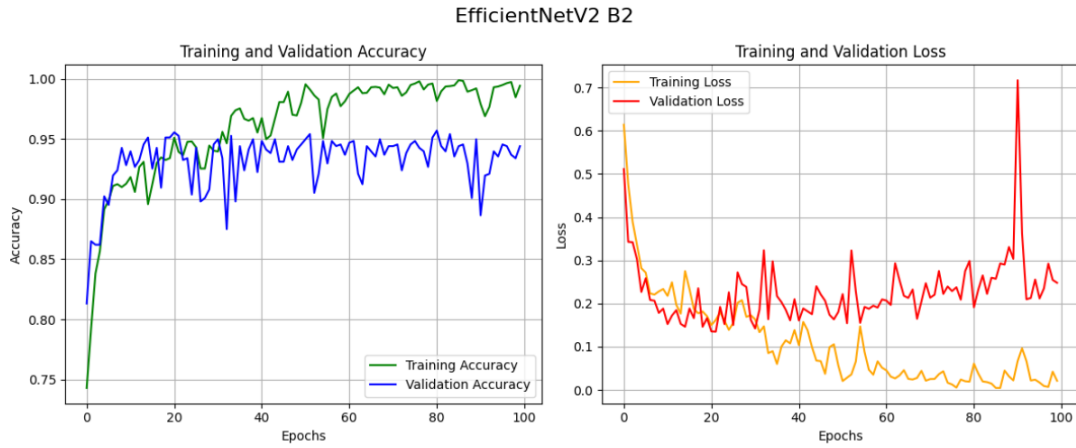


Figure 6.11: Training and validation of EfficientNetV2-B2 on DeepEarthMY Version 2.0 over epochs.

The training and validation accuracy and loss plots reveal distinct learning behaviors between the two dataset configurations. For V1 with random splitting, training loss starts relatively high at around 6.8. Additionally, the validation accuracy exhibits more frequent fluctuations compared to the V2 random split, indicating less stable generalization throughout the training process. In contrast, the V2 random split demonstrates a more stable training process overall, with a smaller spike in loss of 0.7 observed around epoch 90 and no significant accuracy drops throughout training. This indicates that the model trained on V2 demonstrates more consistent learning and better generalization, likely due to the improved distribution of samples in the refined dataset. Overall, these observations indicate that the V2 dataset supports a more stable and reliable learning process, making it better suited for evaluating model generalization performance.

CHAPTER 7

CONCLUSION

In conclusion, the project successfully achieved key milestones, including enhancing the DeepEarthMY dataset with additional annotated images, performing pixel and object count analysis for satellite imagery, and implementing and evaluating multiple deep learning models. These efforts provided valuable insights into image characteristics and dataset distribution.

The dataset analysis reveals significant differences between rural and urban domains in both pixel distribution and object count statistics. Rural areas contain large agricultural fields, wide forests, and water bodies, resulting in dominant pixel counts of Agriculture, Forest, and Water. The urban area has a concentration of Buildings and Road. Moreover, the rural dataset displays larger pixel count fluctuations across Agriculture, Forest, and Water categories than the urban dataset does for Buildings and Roads features. This is due to the rural domain containing extensive landmasses through bigger individual objects such as agricultural territories and extensive water features, while urban regions contain multiple smaller distinct segments of built-up structures

and transportation infrastructure.

During model implementation, we constructed and evaluated the final VMamba model against several established deep learning architectures, including EfficientNetV2, AlexNet, ResNet-50, and the Vision Transformer (ViT). Among these models, EfficientNetV2-B2 achieved the best overall performance across all key metrics, making it the most accurate and reliable model in this study because it balances high precision of 0.8393 and near-perfect recall of 0.9806, resulting in the highest F1-score of 0.9045 and accuracy of 91.00%, while maintaining a relatively low loss of 0.6098, indicating confident and consistent predictions.

After that, the performance of both VMamba and EfficientNet variants were compared. This is done to highlight the strengths of each family. The ultimate goal is to optimize the classification model to accurately distinguish between rural and urban satellite imagery, with enhanced performance and generalization across diverse landscapes. The results show that EfficientNetV2-B2 and VMambaV2-Small performed best in their respective families.

Subsequently, additional experiments were conducted using DeepEarthMY Version 1 (V1) with a random split, in comparison to DeepEarthMY Version 2 (V2)

with both random and manual splits. Interestingly, the model achieved higher accuracy on V1 than on the manually split V2. This difference suggests that the manual split in V2 may have introduced more challenging scenarios or imbalances, potentially reducing model performance. Furthermore, training behavior analysis indicated that the model trained on V2 demonstrated more stable learning dynamics, with fewer fluctuations in loss and accuracy across epochs, implying better robustness and generalization. These findings displays the importance of dataset splitting strategies in achieving both high accuracy and reliable model performance.

While this study has made significant progress in classifying rural and urban satellite imagery using deep learning, there are several areas for future exploration. One potential direction is to further enhance dataset quality by incorporating higher-resolution imagery or additional geographical regions to improve model generalization. Another avenue is to explore advanced data augmentation techniques or semi-supervised learning to address class imbalance and improve performance in low-data regions.

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APPENDIX A

FYP1 Gantt Chart

[illegible]

FYP2 Gantt Chart

[illegible]

APPENDIX B

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 8/11/2025	Meeting No.: 1
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / ~~Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identifying the problem statement
- Defining project goals and objectives
- Outlining project scope and limitations

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning~~ / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Object count of DeepEarth Dataset
- Create a Gantt Chart
- Analysis of pixel count
- Annotation of DeepEarth Images
- Reviewed research papers on satellite image classification methods.
- Compile a list of relevant sources to inform the classification methodology.
- Explore different machine learning techniques for image classification.
- Develop a theoretical framework or justifying the approach with existing theories
- Develop a working prototype
- Write up the background, methodology, results, and analysis

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Start to review DeepEarthMy dataset and related papers



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

Supervisor's Signature



Student's Signature

.....
Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 15/11/2025	Meeting No.: 2
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / ~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identify the problem statement
- Define project goals and objectives
- Outline project scope and limitations
- Review research papers on satellite image classification methods
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~
Requirement Analysis or Theoretical Framework / Design or Research Methodology /
Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Pixel count of DeepEarth dataset
- Object count of DeepEarth Dataset
- Annotation of DeepEarth Images
- Explore different machine learning techniques for image classification
- Compile a list of relevant sources to inform the classification methodology
- Create a Gantt Chart.
- Analysis of pixel count
- Develop a theoretical framework or justifying the approach with existing theories
- Develop a working prototype
- Write up the background, methodology, results, and analysis

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Overwhelming Volume of Studies
Solution: Prioritize studies cited in recent reviews or meta-analyses to identify key contributions.
- Variability in Terminologies
Solution: Focus on extracting common themes and patterns rather than specific terminologies.
- Integration of New Techniques
Solution: Review specialized journals or conferences on geospatial data and machine learning for up-to-date methods.
: Consider including a section on challenges in applying modern techniques, linking it to the need for robust rural-urban classification.
- Complexity in Categorization of Rural Images
Solution: Define objective classification rules based on quantifiable features (e.g., building density, vegetation coverage).
- High Volume of Data:
Solution: Divide the dataset into rural and urban to speed up manual checks.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Good work. Keep up



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
SUPERVISOR'S SIGNATURE
62600 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

.....
Student's Signature

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Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

5. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
6. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
7. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

8. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 22/11/2025	Meeting No.: 3
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identify the problem statement
- Define project goals and objectives
- Outline project scope and limitations
- Review research papers on satellite image classification methods
- Compile a list of relevant sources to inform the classification methodology
- Explore different machine learning techniques for image classification
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review /~~
Requirement Analysis or Theoretical Framework / Design or Research Methodology /
Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Object count of DeepEarth Dataset
- Annotation of DeepEarth Images
- Create a Gantt Chart
- Analysis of pixel count
- Develop a theoretical framework or justifying the approach with existing theories
- Develop a working prototype
- Write up the background, methodology, results, and analysis

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Difficulty in identifying recent and relevant studies on classification methodologies specific to rural, semi-rural, and urban categories.
Solution: Use academic databases like IEEE Xplore, Springer, and ScienceDirect with specific keywords: "rural-urban land classification", and "Satellite image classification"
- Identifying the best machine learning approach
Solution: Selecting an appropriate model architecture to balance accuracy and computational efficiency.
- Missing files in some dataset directories
Solution: Find the updated dataset for DeepEarth

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Learn how to use IEEE Xplorer, Google Scholar, Sciencedirect, etc to search for papers and doing literature review.



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

Supervisor's Signature



Student's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

9. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
10. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
11. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

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12. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 29/11/2025	Meeting No.: 4
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / ~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identify the problem statement
- Define project goals and objectives
- Outline project scope and limitations
- Review research papers on satellite image classification methods
- Compile a list of relevant sources to inform the classification methodology
- Explore different machine learning techniques for image classification
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Analysis of pixel count

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Problem Formulation and Project Planning / ~~Background Study or Literature Review~~ / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Object count of DeepEarth Dataset
- Annotation of DeepEarth Images
- Create a Gantt Chart
- Develop a theoretical framework or justifying the approach with existing theories
- Develop a working prototype
- Write up the background, methodology, results, and analysis

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Please identify suitable additional datapoints to be incorporated in DeepEarthMy dataset



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
... SELANGOR DARUL EHSAN, MALAYSIA
Supervisor's Signature



Student's Signature

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Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

13. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
14. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
15. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

16. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 13/12/2025	Meeting No.: 5
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

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1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identifying the problem statement.
- Define project goals and objectives
- Creating a timeline and milestones.
- Defining project goals and objectives.
- Outlining project scope and limitations.
- Reviewed research papers on satellite image classification methods.
- Compiled a list of relevant sources to inform the classification methodology.
- Explored different machine learning techniques for image classification.
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Analysis of pixel count
- Create a Gantt Chart

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology /~~ Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Developing a theoretical framework or justifying the approach with existing theories
- Developing a working prototype
- Writing up the background, methodology, results, and analysis.
- Object count of DeepEarth Dataset
- Annotation of DeepEarth Images

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

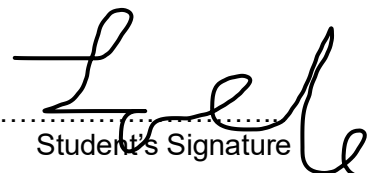
- TIF image files cannot be processed
Solution: Adjust software settings to ensure TIFF support is enabled, or use an alternative program known to handle TIFF images well.
- Out of memory while running MaxVit Transformer
Solution: Start with a smaller batch size (e.g., 444 or 888) and gradually increase it if possible, or use a more powerful machine.
- Input Tensor Shape Mismatch
Solution: Check model input shape, adjust if needed.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) Identify suitable viewer to visualize tif dataset to ensure accuracy.



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

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Supervisor's Signature



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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

17. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
18. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.

19. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

20. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 24/12/2025	Meeting No.: 6
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / ~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identifying the problem statement.
- Define project goals and objectives
- Creating a timeline and milestones.
- Defining project goals and objectives.
- Outlining project scope and limitations.
- Reviewed research papers on satellite image classification methods.
- Compiled a list of relevant sources to inform the classification methodology.
- Explored different machine learning techniques for image classification.
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Analysis of pixel count

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology /~~ Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Developing a theoretical framework or justifying the approach with existing theories
- Developing a working prototype
- Writing up the background, methodology, results, and analysis.
- Object count of DeepEarth Dataset
- Annotation of DeepEarth Images

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- TIF image files cannot be processed
Solution: Adjust software settings to ensure TIFF support is enabled, or use an alternative program known to handle TIFF images well.
- Out of memory while running MaxVit Transformer
Solution: Start with a smaller batch size (e.g., 444 or 888) and gradually increase it if possible, or use a more powerful machine.
- Object count for DeepEarthMY is inaccurate
Solution: Perform a thorough review of a sample of images and annotations to identify patterns in the errors.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

.....
Supervisor's Signature



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Student's Signature

.....
Co-Supervisor's Signature
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.....
Company Supervisor's Signature
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IMPORTANT NOTES TO STUDENTS:

21. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
22. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
23. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
24. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)

Meeting Date: 3/1/2025	Meeting No.: 7
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / ~~Prototype Development or Proof of Concept / Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identifying the problem statement.
- Define project goals and objectives
- Creating a timeline and milestones.
- Defining project goals and objectives.
- Outlining project scope and limitations.
- Reviewed research papers on satellite image classification methods.
- Compiled a list of relevant sources to inform the classification methodology.
- Explored different machine learning techniques for image classification.
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Analysis of pixel count
- Object count of DeepEarth dataset
- Writing up the background study / literature review

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Developing a theoretical framework or justifying the approach with existing theories
- Developing a working prototype
- Writing up the methodology, results, and analysis.
- Annotation of DeepEarth Images

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Boxplot of object count for DeepEarthMY is inaccurate
Solution: further study the detailed object count dataframe, check for any abnormalities
- Summary of literature review is not detailed enough
Solution: Expand the summary and write conclusion from the summary

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

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Supervisor's Signature



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Student's Signature

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Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

25. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
26. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
27. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
28. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2024 (Trimester ID:2430)**

Meeting Date: 10/1/2025	Meeting No.: 8
Meeting Mode: Face-to-Face	
Project ID: FYP01-DS-T2430-0100	Project Type: Research-based
Project Title : Land cover segmentation and classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Sepcialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identifying the problem statement.
- Define project goals and objectives
- Creating a timeline and milestones.
- Defining project goals and objectives.
- Outlining project scope and limitations.
- Reviewed research papers on satellite image classification methods.
- Compiled a list of relevant sources to inform the classification methodology.
- Explored different machine learning techniques for image classification.
- Analysis of DeepEarth dataset
- Check for inconsistencies in DeepEarth dataset
- Categorize DeepEarth images into rural, semi-rural and urban
- Pixel count of DeepEarth dataset
- Analysis of pixel count
- Object count of DeepEarth dataset
- Writing up the background study / literature review
- Developing a theoretical framework or justifying the approach with existing theories
- Developing a working prototype
- Writing up the methodology, results, and analysis.
- Image Annotation Plan for DeepEarthMY dataset

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Details (in point form):

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Input size of images not valid for MaxVit Transformer
Solution: Adjust model input size or resize the images
- Boxplot of object count for DeepEarthMY is inaccurate
Solution: further study the detailed object count dataframe, check for any abnormalities

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

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Supervisor's Signature


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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

29. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
30. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
31. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
32. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 4/4/2025	Meeting No.: 1
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Analysis of custom annotated images
- Added custom-annotated images to the DeepEarthMY dataset
- Analysis of final combined dataset
- Implementation of VMamba
- Testing of VMamba on DeepEarthMY dataset
- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report.
Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee.

This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 11/4/2025	Meeting No.: 2
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images
- Analysis of custom annotated images

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Added custom-annotated images to the DeepEarthMY dataset
- Analysis of final combined dataset
- Implementation of VMamba
- Testing of VMamba on DeepEarthMY dataset
- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature

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Company Supervisor's Signature

(if applicable)

(if applicable)

IMPORTANT NOTES TO STUDENTS:

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2. Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report.
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FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA



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**COMPUTING
& INFORMATICS**

CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 18/4/2025	Meeting No.: 3
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

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1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images
- Analysis of custom annotated images
- Added custom-annotated images to the DeepEarthMY dataset

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Analysis of final combined dataset
- Implementation of VMamba
- Testing of VMamba on DeepEarthMY dataset
- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 25/4/2025	Meeting No.: 4
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images
- Analysis of custom annotated images
- Added custom-annotated images to the DeepEarthMY dataset
- Analysis of final combined dataset

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Implementation of VMamba
- Testing of VMamba on DeepEarthMY dataset
- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



DR. PEE CHIH YANG
ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
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Supervisor's Signature

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Co-Supervisor's Signature
(if applicable)



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Student's Signature

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Company Supervisor's Signature
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IMPORTANT NOTES TO STUDENTS:

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CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 2/5/2025	Meeting No.: 5
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images
- Analysis of custom annotated images
- Added custom-annotated images to the DeepEarthMY dataset
- Analysis of final combined dataset
- Implementation of VMamba

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Testing of VMamba on DeepEarthMY dataset
- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature
(if applicable)

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Company Supervisor's Signature
(if applicable)

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4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6324 Project (FYP2) Meeting Log
Trimester: March 2025 (Trimester ID:2510)

Meeting Date: 9/5/2025	Meeting No.: 6
Meeting Mode: (In-person / Online)	
Project ID: FYP02-DS-T2510-0093	Project Type: Research-based / Application-based / Hybrid
Project Title : Land-Cover Segmentation and Classification of Satellite Images	
Student ID : 1211103024	Student Name: Yap Jack
Student Programme and Specialisation: Bachelor of Computer Science (Hons.) Specialisation: Data Science	
Supervisor Name: Dr. Pee Chih Yang	Co-Supervisor Name: (if applicable) A.P.Dr. Wong Lai Kuan
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Implementation / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion~~

Tasks:

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Create custom annotated Images
- Analysis of custom annotated images
- Added custom-annotated images to the DeepEarthMY dataset
- Analysis of final combined dataset
- Implementation of VMamba
- Testing of VMamba on DeepEarthMY dataset

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Implementation~~ / Testing (Application-based projects) or Evaluation of Findings and Research Contribution (Research-based projects) / Commercialisation Proposal (Application-based projects) or Research Paper (Research-based Projects) / Draft Final Report / Final Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Testing of AlexNet on DeepEarthMY dataset
- Testing of ResNet on DeepEarthMY dataset
- Testing of EfficientNetV2 on DeepEarthMY dataset
- Testing of Vit Transformer on DeepEarthMY dataset
- Compare VMamba with other models
- Draft Final Report

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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ASSISTANT PROFESSOR
FACULTY OF COMPUTING AND INFORMATICS
MULTIMEDIA UNIVERSITY
PESIRAN MULTIMEDIA, 63100 CYBERJAYA
SELANGOR DARUL EHSAN, MALAYSIA

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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature

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Company Supervisor's Signature

(if applicable)

(if applicable)

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4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

APPENDIX C

ORIGINALITY REPORT

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Tesch, Jan Tobias. "Interpretable Deep Learning for Studying the Earth System Soil-Moisture-Precipitation Coupling Across Europe", Rheinische Friedrich-Wilhelms-Universitaet Bonn (Germany)

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Szymon Chmielewski, Andrzej Bochniak, Asya Natapov, Piotr Wężyk. "Introducing GEOBIA to Landscape Imageability Assessment: A Multi-

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Temporal Case Study of the Nature Reserve "Kózuki", Poland", Remote Sensing, 2020

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